

Precision Phenotyping of Type 2 Diabetes in Chinese Populations Using a Variational Autoencoder-Informed Tree Model

Corresponding Author: Professor Jianping Weng

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors have conducted a study developing and comparing a Chinese-specific machine-learning-based predictive model of cardiovascular and kidney outcomes among Type 2 diabetes (T2D) patients to one developed from Scottish data. The manuscript is cumbersome and involves multiple modelling steps, datasets and evaluations, all of which are not adequately described in detail, making the manuscript hard to follow. I find myself with multiple major questions after having read the present manuscript, most of them are outlined below:

1. Abstract: In the abstract it appears as if the variational autoencoder (VAE) informed tree model is solely derived using “ten key clinical features unique to the Chinese population”, when in fact the VAE part of the machine learning pipeline takes in all clinical features.
2. Introduction: Overall, the introduction lacks details on the referenced studies and justification for studies. I find myself asking the following questions: Are the referenced studies using the same input variables? Why does subtyping have limited value when predicting drug response? What is the “Scottish Tree”? Which “golden-standard” methods and imaging are considered in study [12].
3. Introduction: A variational autoencoder is described as a dimensionality reduction technique. While it can be used for dimensionality reduction, it is a probabilistic generative model and it is not guaranteed to provide informative representations for downstream predictive modelling.
4. Methods: The description of the study population is not sufficient. It is described that included individuals are those with T2D within 12 months prior to baseline, where baseline is the first recorded instance of T2D in the database. This makes no sense, as that would imply no T2D events prior to baseline.
5. Methods: There is a lack of justification as to why individuals with major medical conditions were excluded (exclusion criteria (d))
6. Methods: The statement “We collected data needed in validating the Scottish and the Chinese models.” is vague. The authors should be specific – which data (variable names or reference to appendix listing them)
7. Methods: Specify where in the supplementary material the detailed definitions of the diabetes related outcomes are.
8. Methods: Outlier removal and rank normalization is performed without an explicit why (important!) and how (i.e. how was rank normalization in practice performed) and most importantly how many records it affects, and how it e.g. affects the results if it is not considered (as a sensitivity analysis). Adding complex data-preprocessing steps greatly increases model pipeline complexity so these choices should be justified.
9. Methods: There are many examples of sentences where one asks multiple questions regarding justification when reading

it, one example being the sentence “the input features derived from a dataset consisting of 45,589 patients with T2D and 11,340 controls without diabetes, which was sourced from the CRDS database by random selection of 10% of the participants and exclusion of those who had important data missing”. What input are we talking about? Why are you considering controls and why this ratio? What is considered ‘important’ data to be missing?

10. Methods: A Variational Autoencoder (VAE) is introduced as a feature extractor. However, a justification as to why this is needed is lacking. While a VAE can be used as a feature extractor, it is a generative model partly trained to reconstruct its input. Its introduction as a feature extractor adds layers of complexity to the model pipeline such as: architecture design in terms of how many layers to include, regularization, latent dimensionality, training parameters (i.e. dropout, learning rate, batch size, etc.) – these details are also not given in detail nor justified in the manuscript. There are multiple machine learning methods perfectly capable of handling large feature inputs such as e.g. the LASSO or Elastic net. Investigations on whether simpler approaches would work are needed as the simplest possible method that fits the data well should be promoted.

11. Methods: There is a lack of detail in the description of the methods to a point where reproducing the experiments are impossible. The number of cross validation folds are not specified, and libraries of the machine learning algorithms considered are not specified, nor definitions of their hyperparameter spaces.

12. Methods: It is claimed that the GBC method achieves “highest accuracy” and therefore is chosen as a “final model”. However, the authors are considering a multitude of outcomes so is that really the case for ALL of these? In addition, is accuracy really the best measure to consider? One could imagine that some of the considered outcomes are rare leading to class imbalance. Alternative measures not so sensible to class imbalance (e.g. ROC or PR AUC) should at the very least be considered for model selection, and at the very least reported. Further – there is no mention of how the prediction threshold was decided.

13. Methods: It is not specified if both the mean and variance of the approximate posterior gaussian are used as extracted features in the VAE pipeline, or simply the mean.

14. Methods: The feature importance method is not adequately described.

15. Code availability: Given the lack of detail in the description of the methods and no code availability it is impossible to reproduce the presented results of the manuscript, even if one would have access to the data.

Reviewer #2

(Remarks to the Author)

In this manuscript, the authors sought to characterize Chinese-specific T2D heterogeneity using deep learning-informed tree models. The authors have performed novel analyses, compared their findings to a previously published Scottish study by Nair et al., and validated their findings in multiple sub-cohorts. Their study provides a deeper understanding of the phenotypic variations in T2D in the Chinese population.

I think the unique risk factors for diabetes in the Chinese population should be mentioned in the introduction from the available literature. This will help clarify why this study is justified. Although over 8.6 million individuals are in the cohort, only a part of it is used in the analysis. The sentence might be misleading to the readers. In the introduction, 2nd paragraph, the second sentence, “we come,” may be removed. I am not sure whether clustering is an arbitrary allocation. Even clustering methods utilize information from the input clinical parameters, while I do agree that discretization is of limited utility.

In the CRDS cohort, the number of participants with long-term follow-up was relatively low. How might this have affected the tree structure and deep learning during training? Also, please specify how many participants had completed 10-year follow-up data in the text when you first described it. I assume this proportion would be lower given that only 5.6% of T2D patients had 5 years of follow-up initially. However, I later see that the number of long-term follow-ups has increased, which introduces some confusion. It would be helpful for the authors to clarify this discrepancy. Additionally, how might this variation in follow-up duration affect the algorithm’s ability to learn patterns effectively, particularly for predicting long-term outcomes such as diabetic retinopathy?

The observation was that HbA1c had modest effect, while lipids had the most substantial effect. This is surprising, as one would expect HbA1c to be the strongest driver when studying diabetes outcomes. Also, liver disease exerts an undue influence on specific phenotypes such as DR and CVD. I think this is an artifact arising from the huge burden of liver disease in the Chinese population (PMID: 25164003). However, I am not denying that it could indeed have a significant influence on diabetes and its outcomes. Therefore, the authors must consider reporting the burden of liver diseases in the sample for each diabetes complication explored in this manuscript as a table.

BMI differences between ethnicities could introduce specific errors, especially in cases like this, where there is evidence of interplay of BMI with other risk factors. Does this also imply that the effect of BMI on diabetes phenotypes may have been masked due to the mismatch? The authors have not stated explicitly in the text, or supplementary which variables were considered for selecting important features. In the following line, BMI was repeated in the text.

Predicting the long-term risk of diabetic retinopathy (DR) remains challenging when relying solely on traditional risk factors (Figure 1). A higher risk for diabetic retinopathy (DR) was observed in individuals with high HDL-C and indicators of liver function. However, these findings are not well-supported by existing large-sample studies in the Chinese population. Furthermore, high HDL-C levels are generally considered protective against diabetic retinopathy (DR) due to their anti-inflammatory properties. However, conflicting findings—primarily from cross-sectional studies—do exist. The authors should acknowledge these discrepancies and offer a possible explanation for the observed association in the discussion, supported by appropriate references. Genetics alone cannot account for such heterogeneity, particularly when the phenotype associations come into question.

What outcomes were different between CRDS and JADE? Please add to the main text. More Scottish participants shifted their position after 5 years compared to the Chinese participants. Again, we would expect that the Asian population is likely to be more vulnerable. What could be the reason behind this finding?

Was any data curation performed for the phenotypes? None has been mentioned except outlier identification and corresponding ICD codes. What retinopathy grades were included in the analysis (e.g., proliferative DR, any DR, etc.)? The phenotype associations are likely to change as per the DR definitions. The VAE model was trained on a dataset in which the number of controls was four times smaller than the number of cases. How might this imbalance affect the model's learning accuracy? So, does this mean the data was trained on more than 56,000 individuals? Is that number mentioned in the text, and is it consistent throughout?

I am looking at Figure 1 and would have expected a higher number of DR cases—only 8.6% after 5 years seems surprisingly low. Normally, one would see more than 20%. The dimension 2 is associated with obesity and lipids. That's interesting!

There are numerous figures and frequently changing numbers throughout the manuscript, which makes it challenging for the reader to follow the results. The authors may consider consolidating them to improve clarity and interpretability.

Reviewer #3

(Remarks to the Author)

In their manuscript, the authors used a novel two-step approach—first validating a previously published 'Scottish tree' model in a Chinese population, then constructing a Chinese-specific phenotyping tree using a variational autoencoder (VAE)-based feature selection framework.

This manuscript presents a comprehensive and innovative approach to characterizing the heterogeneity of T2D in a large Chinese population using a deep learning-based phenotyping framework. By first validating the applicability of a previously published 'Scottish tree' model and then constructing a Chinese-specific tree based on VAE-based feature selection, the authors provide compelling evidence for ancestry-specific disease patterns of diabetes and complication risks. The study leverages a massive electronic health record dataset, robust statistical models, and external validations across independent cohorts, significantly enhancing the credibility and generalizability of the findings.

Major comments:

1. 'The data included all the patients whose data is accessible from the participating centers' medical record systems.' Were all the data during the period accessible in those centers? If not, what are the percentages of accessible data, and what determined the accessibility? In addition, it is good to know the distribution of these 24 medical centers.
2. The rationale for selecting exactly ten features in the final model is not clearly explained. Why was this number chosen? It would be helpful if the authors could clarify whether they tested different numbers of input features and how this affected the tree structure, complication risk gradients, or model performance.
3. Could you explain the X and Y axes of the DDRTree in the Methods section and the linear regression analyses of each phenotype and outcome against the tree dimensions?
4. As HbA1c is a key indicator for T2D diagnosis and monitoring, its low importance in the model could be discussed further. The authors may consider whether this reflects ethnic differences or potential biases.
5. Individuals shifted their positions after five years. It may be helpful to conduct a supplementary analysis to examine whether those who moved into the high-risk group indeed experienced more complications.
6. Could the authors explain how the influence of the variables on the tree coordinates was determined in the 'development and performance of the Chinese tree model' section (Figure 2D)?
7. The authors used Fine-Gray model and sub hazard ratios (sHRs) with death as a competing event for the main analysis instead of a sensitivity analysis. Please elaborate.

Minor comments:

8. 'For this study, we included individuals aged 18 years or older from the CRDS who were diagnosed with type 2 diabetes (T2D; ICD-10 code E11) within the 12 months prior to baseline. Baseline was defined as the first recorded instance of a T2D diagnosis in the database.' It is a bit confusing about the time for baseline and T2D diagnosis.

9. Please use JADE cohort throughout the manuscript and the supplement instead of Hong Kong dataset in some of the tables and figures.

10. Figure S15 was not mentioned in the manuscript.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Overall summary

The authors have made substantial changes to the present manuscript. The authors, in addition, tried to address the feedback provided. However, in some cases the questions raised by the previous reviewers have not been directly addressed. In addition, the extensive revisions have also raised new questions and, in some cases, created additional issues that need clarification.

Therefore, my revised revisions are structured in the following way. First, general concerns, followed by a point-to-point evaluation of the responses to my previous comments (numbered from 1 to 15). Then, new comments based on the direct revisions provided from the first version of the paper to the current one, numbered from 16 to 45.

General concerns

Despite substantial revisions, the manuscript remains difficult to evaluate due to unresolved methodological ambiguity and new inconsistencies introduced during revision. My overarching concerns are:

1) Coherence and consistency

The described pipeline, figures, and methods do not align (e.g., discrepancy between Fig. 1 and the narrative about how VAE outputs feed downstream steps; unclear group definitions in Fig. 6f; table totals that exceed the stated number of outlier individuals). Several minor errors (typos, repeated sentences, and a mislabeled loss term) further complicate interpretation (see comments 2, 8, 26, and 39–45).

2) Role and justification of the VAE

It remains unclear whether the VAE is used for representation learning, feature selection, denoising, or all three, and why this choice is preferable to simpler alternatives (e.g. a vanilla autoencoder, PCA, or purely supervised selection). Assertions about “informative” latent structure and “denoised input” are not supported by comparative analyses (see comments 3, 10, 14, 18, 27–28, 30–33, and 36).

3) Evaluation design, hyperparameters, and reproducibility

The manuscript lacks essential details on hyperparameter search spaces, selection criteria, and safeguards against data leakage. Model performance is reported without clear nested CV, fold definitions. It is also unclear when models use the full feature set versus the 10 “VAE-selected” features (see comments 10–12, 25–26, 33–35, 38). The proposed “stability” metric based on cosine similarity across independently trained VAEs is not valid given rotational/permutation of latent spaces (see comment 24).

4) Use of controls and inclusion/exclusion criteria

The inclusion of non-diabetic controls in representation learning is not sufficiently justified if the stated goal is to extract diabetes-specific patterns. The chosen case:control ratio is also not motivated. On the other hand, exclusion of multimorbid individuals may limit real-world applicability and should be addressed (see comments 5, 9 and 21).

5) Claims that exceed current evidence

Statements about longitudinal “phenotypic shift,” superiority of DDRTree over traditional clustering, mitigation of confounding via VAE-based selection, clinical relevance of the continuous model, and “early interventions” are insufficiently supported by statistical testing, benchmarking, or appropriate references (see comments 16–17, 20, 28–29, and 41–42).

Point-by-point evaluation of responses

Comment 1: In my previous comment #1 I addressed inconsistencies in the model pipeline in the abstract. The authors have revised this part adequately.

Comment 2: The authors addressed my previous comment #2 by providing more context in the introduction. However, their introduced Figure 1 does not agree with the methodology described elsewhere in the manuscript. In particular, in “STEP2: Selecting T2D Features”, it appears if the reconstructed input from the variational autoencoder (VAE) is fed into the machine

learning selection pipeline. Instead it should (probably?) reflect that the VAE is only used for feature selection based on feature attribution to the latent representations.

Comment 3: In my previous comment #3 I noted the authors that the VAE is a generative model, not necessarily guaranteed to provide informative representations for predictive modelling. The authors replied that in fact it is solely used for feature extraction, and that its “design allows for a continuous and smoother latent space, making VAE not only more generative in nature, but also better suited to capture the intrinsic structure of the input data”. When making these statements regarding the latent space, it appears as if the authors compare the VAE approach to other dimensionality reduction techniques (such as e.g. a vanilla AE, a denoising AE, PCA, or other generative models), but this is not the case. Following, the authors write in their response “While reconstruction is the primary training objective, we interpret this bottleneck as a form of feature-preserving compression, where essential input characteristics are retained to ensure accurate reconstruction. As such, the reconstructed input (X') remains close to the original input (X), validating that the learned representations preserves relevant information”. There are two things to this comment. Firstly, a good reconstruction does not validate that the representations preserves ‘relevant’ information. The VAE can simply overfit to a ‘dictionary’ remembering the input data from the latent representations, without the latent representations having any meaning (see e.g. Alain, Guillaume, and Yoshua Bengio. “What regularized auto-encoders learn from the data-generating distribution.” *The Journal of Machine Learning Research* 15.1 (2014): 3563-3593.). Secondly, given the authors focus on the reconstruction error, why not just use the (much simpler) vanilla autoencoder? Why add the complexity of the VAE? Later in the authors response they touch upon the nonlinear capabilities of the VAE, which are desirable for modelling complex correlations, but in no way does it guarantee “denoised input” or “informative patterns”.

Comment 4: The authors clarified and addressed my previous comments adequately.

Comment 5: My previous comment #5 addressed a lack of justification for individuals with major medical conditions being excluded. The authors responded saying the excluded to “reduce potential confounding effects that could distort the identification of T2D subgroups”. It is true that individuals with many complicated comorbidities would likely alter the biomarker profiles used in the clustering framework. On the other hand, many individuals living with diabetes are multimorbid (often upwards of 33%, see ref Khunti, K., Chudasama, Y. V., Gregg, E. W., Kamkuemah, M., Misra, S., Suls, J., ... & Valabhji, J. (2023). Diabetes and multiple long-term conditions: a review of our current global health challenge. *Diabetes Care*, 46(12), 2092-2101), so it is true that inclusions of comorbidities could bias diabetes-specific phenotypic patterns, but it also excludes potential usefulness of the derived model in the real world where multimorbid diabetes patients are prevalent.

Comment 6: The authors clarified and addressed my previous comments adequately.

Comment 7: The authors clarified and addressed my previous comments adequately.

Comment 8: My previous comments address the complex data preprocessing steps the authors apply in their modeling pipeline. The authors responded with clarification on the steps. In particular, the authors provide a table with individuals with outlier data (in total 575 individuals), but the numbers in the table are larger than 575 so I am unsure what to make of the table. Could the authors please clarify?

Comment 9: My previous comment 9 raised concerns regarding why controls were sampled for the representation learning part of the model pipeline. The authors replied: “The inclusion of controls was intended to enhance the contrast between diabetic and non-diabetic phenotypes, allowing for better identification of diabetes-specific patterns, particularly during unsupervised learning and latent space evaluation”. I do not believe that answers my question. Can the authors elaborate? Adding the non-diabetic individuals is rather counterintuitive IF the goal is diabetes-specific patterns. So-called “enhanced contrasts” will distort the diabetes-specific patterns, not enhance them. If applying the authors logic then I see no reason to exclude multimorbid diabetes patients, as those could then also “enhance contrasts”.

Comment 10: In my previous comment #10, I urged the authors to justify the choice of VAE, as opposed to simpler alternatives. The authors have attempted to adhere to my comments by adding a LASSO and elastic net model (showing the performance of the model), yet it is unclear how the authors implemented these models. Did they let apply them on the full dataset or the subset of 10 features as the other ML model (i.e. the GBC). In general, it would be useful to compare the performance of the ML models when using the 10 “VAE-selected” features versus using the full featureset. Also, the manuscripts remains with no details regarding which hyperparameters were chosen in the supervised machine learning models. For example, in the LASSO model the lambda parameter controls the L1 shrinkage penalty. Ideally, these should be tuned, and the total performance evaluated in a nested CV setup.

Comment 11: My previous comment #11 addressed the lack of detail in describing the methods. The authors have done some work to improve the manuscript in this regard, but there are still details and justifications lacking (see additional comments)

Comment 12: My previous comment #12 concerned the reporting of ML performance and the metrics used. The authors have partly addressed these comments by including the ROC AUC and PR AUC for their assessment, yet I wonder how hyperparameters were chosen when making these assessments (see comment 10).

Comment 13: The authors clarified and addressed my previous comments adequately.

Comment 14: My previous comment #14 concerned the feature importance measure based off the VAE weights. The authors

have partly addressed these comments. Yet, the method is still poorly presented mathematically in the methods section. The authors write in their reply: "A weight-based approach, computing the sum of absolute encoder weights from each input to all hidden nodes". I wonder if this applies to both the means and the variances in the hidden layer, or solely the means?

Comment 15: The authors clarified and addressed my previous comments adequately.
New comments based on revisions made.

Major comments

Comment 16: In the abstract, a "significant longitudinal phenotypic shift" is described. Yet, in the manuscript no formal statistical tests are performed. Even if performed it is difficult to translate those into any meaning, as they are based on latent dimensions from a dimensionality reduction method.

Comment 17: In the introduction, the authors have added the sentence "This flexible framework has also been successfully applied to type 1 diabetes and prediabetes, where it enabled the delineation of heterogeneous disease trajectories and facilitated the identification of early interventions tailored to specific risk patterns", followed by two references. The two referenced studies do not directly facilitate nor evaluate so-called "early interventions". They simply briefly discuss that such a model could be used for design of such interventions. There is a fundamental difference between using risk prediction models for providing information for e.g. a GP and risk prediction models designed specifically with an intervention in mind (see e.g. van Geloven, Nan, et al. "The risks of risk assessment: causal blind spots when using prediction models for treatment decisions." *Annals of internal medicine* (2025)).

Comment 18: In the introduction, the authors have added the sentence: "While the learned features are not specifically optimized for downstream predictive modeling tasks, they can still reveal meaningful patterns and help identify clinically relevant variables when conventional selection methods are insufficient." While this is true, the authors have not argued for why these conventional selection methods are insufficient in their setting, nor have they looked at e.g. differences in selected variables if one employs a "traditional" selection method, compared to their complex VAE-based setup.

Comment 19: In the "Mapping the Chinese individuals onto the Scottish tree model" section, the authors write "a distance-based algorithm assigned each individual to the nearest node on the Scottish reference tree." The authors should be specific in which algorithm (or metric) is used.

Comment 20: In the same section, the authors have now clarified "For each outcome, we applied Fine–Gray competing risk models, using death as a competing event. Ten-year event probabilities were estimated for each individual based on their coordinates in the two-dimensional tree space." Were these predictions based solely on the coordinates? Why not control for sex and age, to ensure that differences in estimates are not driven entirely by differences in baseline characteristics by the patients.

Comment 21: In the "Development and performance of the Chinese tree model" section, the authors write "The VAE model was trained using data from 45,589 individuals with type 2 diabetes (T2D) and 11,340 healthy individuals (without diabetes)." This choice (both in terms of why healthy individuals are included, but also the ratio of T2D to controls) should be justified, and it is not currently so in the present manuscript.

Comment 22: In the same section, the authors mention "we evaluated both the reconstruction accuracy of the input data and the model's stability across multiple refitting procedures". It is not clear what is meant by "the model's stability here".

Comment 23: In the same section, the authors mention a reconstruction accuracy ranging from 0.85 to 1.0 which seems very high. A perfect reconstruction for a VAE could imply that a dictionary mapping was learned, and that the KL penalty factor in the ELBO were not enforced. How do the KL divergence loss term look in these cases? Have you visualized the latent representations for these "perfect" situations to ensure you have not just learned a dictionary.

Comment 24: The authors use average change in cosine similarity "within the latent space" as a measure of stability. However, VAEs don't have an inherent "coordinate system" in the latent space. If you train two VAEs on the same data (or different folds), their latent axes can rotate, flip, or permute without changing the learned distribution. So the same data point may map to quite different coordinates in the two spaces. That means cosine similarity between the raw latent vector of VAE A and VAE B is not guaranteed to reflect meaningful closeness.

Comment 25: In the same section, the authors write "We first trained several models using all input features and identified the Gradient Boosting Classifier (GBC) as the best-performing model based on 10-fold cross-validation." How were hyperparameters for the GBC (and other algorithms) chosen in this evaluation setup? It is not clear if an internal optimization of hyperparameters was performed.

Comment 26: The authors write "the accuracy had already exceeded 0.915, closely approaching the performance of the full-feature model (Supplementary Fig. 5)." The full-feature model is not in supp fig 5, so it would be useful to contrast the performance of the ML algorithm based on the subset and based on the full features. This goes in general for all of the evaluated models.

Comment 27: In the discussion the authors write "We then applied gradient-boosting classifiers to identify the top variables contributing to subtype discrimination." This is confusing as it is not what is described in the rest of the manuscript. My

understanding is that you selected features based on reconstruction importance in the VAE and/or shap values linking to their representations.

Comment 28: The authors claim that "Importantly, this method helps mitigate the confounding effects of variable co-dependence, which may bias subtype classification in traditional models relying on manually selected features." How exactly is this the case? A data-driven selection (via VAE) is less subjective than picking only "clinically relevant" variables – I agree with that. However, VAEs don't remove variable codependence; they model it. The latent factors will still be shaped by correlated inputs. The choice of "most influential variables" is model-dependent (different seeds, hyperparameters, or architectures could change the ranking). Influence measures can still be biased toward variables with strong variance or measurement stability, not necessarily those with the deepest clinical meaning. Saying that the selection process mitigates confounding effects is too strong. It mitigates selection bias, but not confounding directly.

Comment 29: The authors write: "This large sample size enabled us to construct a robust two-dimensional tree structure using the DDRTree algorithm, which outperforms traditional clustering methods in capturing the complexity and continuity of T2D progression." This is a claim. The authors have not benchmarked against traditional clustering methods nor referenced to studies who do.

Comment 30: In the "Design of VAE section" the authors write that the mean and log variances projected by the encoder defines a multivariate Gaussian distribution, but then they write $N(0,1)$, suggesting it be a multivariate standard normal. If so, how is the information from the encoder used? Based on the current description it is not clear.

Comment 31: In the same section – the authors write that they use both ReLU and LeakyReLU as activation functions – could the authors clarify which one, and if both describe in which settings one is used instead of the other?

Comment 32: In the same section it is described that a flat 10% dropout and 32-dimensional latent space was used, yet you write in the manuscript (and previous rebuttal letter) that you optimized over these.

Comment 33: In the "model training" section, the authors mention the evidence lower bound (ELBO), yet it is not described elsewhere in the manuscript. It's relation to the loss function in the manuscript should be specified.

Comment 34: In the hyperparameter optimization section, it is stated that the dataset was split 90/10. Since the authors are using this to guide model selection, I suppose this split was performed on the train dataset?

Comment 35: The authors write that performance was evaluated using both test log-likelihood and reconstruction accuracy. How was test log-likelihood assessed? It is not clear in VAEs how to do this.

Comment 36: The formula for the feature importance implies a sum over weights in just a single hidden layer, yet the authors also evaluate architectures with more than 1 hidden layer before the latent bottleneck. Can you generalize the formula to these cases as well.

Comment 37: The authors write "The difference in model performance was assessed as the absolute differences of the latent representation when changing each input to missing for all individuals and passing it through the trained model." How did you in practice change the input to missing? It is not clear from the text.

Comment 38: In the "Comparison for feature selection and model performance" section, it is described that a number of ML algorithms were compared using 10-fold CV, and the 10 clinical features. However, it is not described if/how models utilizing the full featureset perform.

Comment 39: In Figure 6, panel f, a ribbon chart is showing transitions between groups. These groups appear on the right in the figure, but it is not specified how these were defined. Also, there are some individuals not belonging to a group – is this intended? The methods section lacks details on how these were defined.

Minor comments

Comment 40: in the abstract, there is a missing "the" in the sentence "built a tree structure using Discriminative Dimensionality Reduction Tree (DDRTree) algorithm."

Comment 41: The authors write "In the Chinese tree, 6.45% (95 out of 1,472, from 319 26,000 Chinese individuals) showed a shift (distance > 1) from their baseline position." How is this distance calculated?

Comment 42: In the discussion, the authors write: "This continuous model structure may offer higher clinical relevance than rigid cluster boundaries, especially for diseases like T2D where pathophysiology lies on a continuum." There are ways to enforce non-rigid cluster boundaries in cluster algorithms – see e.g. fuzzy clustering methods or probabilistic clustering.

Comment 43: In lines 600-601 you write first decoder and then encoder. I suppose the encoder should be changed to decoder?

Comment 44: There is an error in the loss formula – MCE is written instead of MSE.

Comment 45: There are repeat sentences in lines 630-632.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

Dear Authors,

Thank you for your revisions. I have reviewed your responses and am pleased to inform you that all the comments from my side have been addressed satisfactorily.

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

The authors addressed my comments adequately. I have no further comments.

(Remarks on code availability)

Version 2:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors have addressed my previous comments. My only remaining comments are minor and consist of typos I identified while going through the new manuscript.

Specifically:

- 1) In figure 1 and the panel of step 2, the word "hyperparamter" is written instead of hyperparameter.
- 2) In line 672, it reads: "where corresponds to the inverse of C". I believe that this should read "which corresponds to the inverse of C".

(Remarks on code availability)

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Point-by-point response to the reviewers' comments

Reviewer #1 (Remarks to the Author):

The authors have conducted a study developing and comparing a Chinese-specific machine-learning-based predictive model of cardiovascular and kidney outcomes among Type 2 diabetes (T2D) patients to one developed from Scottish data.

The manuscript is cumbersome and involves multiple modelling steps, datasets and evaluations, all of which are not adequately described in detail, making the manuscript hard to follow.

I find myself with multiple major questions after having read the present manuscript, most of them are outlined below:

1. Abstract: In the abstract it appears as if the variational autoencoder (VAE) informed tree model is solely derived using “ten key clinical features unique to the Chinese population”, when in fact the VAE part of the machine learning pipeline takes in all clinical features.

Response: We appreciate your careful reading and insightful comment. We acknowledge that the abstract may have caused confusion regarding the role of the VAE in our analysis.

To clarify, the VAE model in our pipeline was not directly used to construct the tree model itself, but rather to identify the most informative clinical features to be included in the subsequent tree model. In contrast to previous studies that selected variables based mainly on domain knowledge or correlation analyses without a formal variable selection procedure (*Lancet Diabetes Endocrinol* 6, 361-369 (2018) or *Nat Med* 28, 982-988 (2022) or *Cell Rep Med* 6, 102212 (2025)), we aimed to adopt a data-driven approach to minimize potential bias or redundancy in feature selection.

Specifically, the VAE was trained using all available clinical features, and it learned a latent representation of the data. Based on the contribution of each variable to this latent space, we ranked the variables and selected the top 10 features that best captured the underlying clinical heterogeneity. These selected features were then used to construct the Chinese-specific tree model via the Discriminative Dimensionality Reduction Tree (DDRTree) algorithm.

We have revised the **Abstract** in Page 3: “To capture Chinese-specific heterogeneity, we applied a variational autoencoder (VAE) and machine learning framework to identify key clinical features and built a tree model using Discriminative Dimensionality Reduction Tree (DDRTree) algorithm” accordingly to more clearly reflect this process and prevent misunderstanding.

2. Introduction: Overall, the introduction lacks details on the referenced studies and justification for studies. I find myself asking the following questions: Are the referenced studies using the same input variables? Why does subtyping have limited value when predicting drug response? What is the “Scottish Tree”? Which “golden-standard” methods and imaging are considered in study [12].

Response: Thank you for your question. Recent advances in data-driven disease stratification have demonstrated that the clinical heterogeneity of T2D can be captured using a tree-like graph structure derived from unsupervised manifold learning. Specifically, *Nair et al.* applied the reversed graph embedding-based DDRTree algorithm to nine readily available clinical variables—HbA1c, BMI, TC, HDL-C, ALT, Cr, SBP, TRIG, and DBP—in a large Scottish cohort of T2D patients. This “Scottish Tree” model enables each individual with T2D to be projected onto a low-dimensional tree-shaped manifold, in which their position correlates with long-term risk of multiple diabetes-related complications including microvascular and macrovascular diseases. A subsequent validation study mapped deeply phenotyped patients from the German Diabetes Study (GDS) and the LURIC cohort onto the Scottish Tree to explore the

biological underpinnings of the tree structure. This study used gold-standard methods including hyperinsulinemic-euglycemic clamps for assessing insulin resistance, intravenous glucose tolerance tests for measuring β -cell function, liver fat, and heart failure subtypes to demonstrate that key pathophysiological axes such as insulin sensitivity, adiposity, and ectopic fat deposition vary across the tree. These findings support the utility of the Scottish Tree model as a biologically and clinically meaningful framework for representing heterogeneity in T2D. The tree-based manifold model provides a continuous, spatially-informed representation of patient states, potentially offering more refined insights into risk and therapeutic targeting.

In previous studies such as *Ahlqvist et al.*'s landmark classification of T2D subtypes (SIRD, SIDD, MOD, MARD, SAID), the choice of input variables—glutamate decarboxylase antibody (GADA), age at diagnosis, BMI, HbA1c, HOMA2- β (β -cell function), and HOMA2-IR (insulin resistance)—was largely based on clinical expert consensus and existing biological knowledge (*Lancet Diabetes Endocrinol*, 6 (5) (2018), pp. 361-369). Similarly, the nine clinical variables used to construct the original Scottish Tree model were also selected based on prior assumptions about their relevance to T2D pathophysiology and outcomes.

In our initial analyses, we replicated the Scottish Tree on the Chinese T2D population to assess the generalizability and structural consistency, following the approach of the LURIC cohort mapping study (*Lancet Diabetes Endocrinol*. 2024 Feb;12(2):119-131). However, we also recognized that variable selection based solely on expert opinion may overlook patterns that are more objectively informative in different populations. In response, we adopted a data-driven strategy combining machine learning and deep learning to systematically identify features most predictive of T2D classification within the Chinese population. This process yielded ten variables, of which eight overlapped with those in the Scottish Tree, while two additional features (Heart rate and LDL-C) were found to contribute meaningfully to classification performance. These added variables may capture unique aspects of T2D heterogeneity in Chinese populations.

To address this, we revised **Introduction section** and **Figure 1** to better illustrate the study design pipeline, including the rationale and results of feature selection. We hope this clarifies the basis for our final model design and demonstrates the added scientific value of our approach in adapting the existing tree framework to better fit the characteristics of the Chinese T2D population.

3. Introduction: A variational autoencoder is described as a dimensionality reduction technique. While it can be used for dimensionality reduction, it is a probabilistic generative model and it is not guaranteed to provide informative representations for downstream predictive modelling.

Response: We appreciate your insightful comment and agree that the VAE is primarily a probabilistic generative model, rather than a conventional dimensionality reduction technique. We acknowledge that our initial description may have oversimplified the role of VAE in this context.

In response, we have revised the **Introduction** to clarify that VAE learns a probabilistic latent representation of high-dimensional data and can be used for dimensionality reduction by capturing non-linear structures in the data. While it is true that the latent features derived from VAE are not explicitly optimized for downstream predictive modeling, in our study, we used the VAE representation primarily for variable ranking and selection purposes. This was intended to provide a data-driven approach to identify the most informative clinical features, as opposed to relying solely on expert-driven or correlation-based feature selection as done in prior studies.

Moreover, although VAE originates from the autoencoder framework, its key distinction lies in modeling the latent representation as a probability distribution, characterized by learnable means and variances. This design allows for a continuous and smoother latent space, making VAE not only more generative in nature but also better suited to capture the intrinsic structure of the input data(<https://doi.org/10.48550/arXiv.1312.6114>).

While reconstruction is the primary training objective, we interpret this bottleneck as a form of feature-preserving compression, where essential input characteristics are retained to ensure accurate reconstruction. As such, the reconstructed input (X') remains close to the original input (X), validating that the learned representation preserves relevant information. Despite its generative formulation, the VAE thus provides a latent representation that captures the most informative and stable components of the original input features, which is particularly beneficial when clinical data is noisy or contains complex correlations that are not well modeled by linear methods. Specifically, given that clinical variables often have moderate dimensionality but may contain measurement noise or hidden confounding (*Nature*. 2021 Nov;599(7883):91-9), the VAE's generative structure was leveraged to denoise the input and highlight informative patterns. This is a strategy increasingly used in omics research (*Nat Biotechnol* 41, 399-408 (2023)), where unsupervised learning models help extract stable and biologically relevant features from complex datasets. Compared to expert-driven or correlation-based feature selection methods used in previous tree-based studies, our approach provides a more systematic and data-driven basis for identifying clinically meaningful variables.

This rationale underpins our use of the VAE not for generation purposes, but as a robust tool for feature denoising and selection, ultimately enabling a more data-driven and potentially generalizable approach to identify key clinical markers for T2D stratification in the Chinese population.

We have updated the manuscript accordingly to ensure a more accurate and nuanced description of the role of the VAE and its relevance to our analysis.

4. Methods: The description of the study population is not sufficient. It is described that included individuals are those with T2D within 12 months prior to baseline, where baseline is the first recorded instance of T2D in the database. This makes no sense, as that would imply no T2D events prior to baseline.

Response: We thank you for pointing out this lack of clarity. We agree that the original description may have been misleading. In our study, we aimed to include individuals with newly diagnosed T2D to better capture early disease features and reduce the confounding effects of long disease duration, and to use their data closest to their diagnosis for subtyping. We defined “newly diagnosed T2D” as T2D diagnosed for no more than 12 months. To achieve this, we identified individuals, whose electronic health records documented a first-ever recorded diagnosis of T2D (ICD-10 code E11) as the baseline, and then reviewed their medical notes and prescription histories to confirm that the diagnosis of T2D was made within the preceding 12 months. Therefore, the phrase "diagnosed within the 12 months prior to baseline" means cases where baseline was the first documented diagnosis of T2D in the database, and the overall disease duration was ≤ 12 months, based on clinical documentation.

We have revised the **Methods section** to clarify this process accordingly (Page 21, Lines 515-518).

5. Methods: There is a lack of justification as to why individuals with major medical conditions were excluded (exclusion criteria (d))

Response: We appreciate your observation. Individuals with major medical conditions, such as malignancy, organ transplantation, or end-stage renal disease were excluded to reduce potential confounding effects that could distort the identification of T2D subgroups. These conditions often involve complex metabolic disturbances, intensive treatments, and high comorbidity burdens, all of which may independently alter clinical features and biomarker profiles used in our clustering framework. Including such individuals could have biased the modeling of diabetes-specific phenotypic patterns. We have now added this justification to the **Methods section** with highlight (Page 21, Lines 521-523).

6. Methods: The statement “We collected data needed in validating the Scottish and the Chinese models.” is vague. The authors should be specific – which data (variable names or reference to appendix listing them)

Response: Thank you for pointing this out. We agree that the original statement lacked specificity. In our analysis, we extracted a comprehensive set of demographics, clinical, laboratory, and treatment-related variables to support model validation. These variables included anthropometric measures, glycemic and lipid markers, renal and liver function indices, hematologic parameters, comorbidities, and medication use. A full list of the variables used is now provided in the revised Methods section and has been summarized in **Supplementary Table 2** for clarity and reproducibility.

We have updated the manuscript accordingly to ensure transparency (Pages 23-24, Lines 599-601).

7. Methods: Specify where in the supplementary material the detailed definitions of the diabetes related outcomes are.

Response: Thank you for your comment. We have clarified the location of the detailed definitions of diabetes-related outcomes in the revised manuscript. These definitions are provided in **Supplementary Table 1**. We have updated the **Methods section** accordingly to reference this table (Pages 22-23, Lines 545-585).

8. Methods: Outlier removal and rank normalization is performed without an explicit why (important!) and how (i.e. how was rank normalization in practice performed) and most importantly how many records it affects, and how it e.g. affects the results if it is not considered (as a sensitivity analysis). Adding complex data-preprocessing steps greatly increases model pipeline complexity so these choices should be justified.

Response: We appreciate your important comment. We have revised the Methods section to explicitly clarify the rationale and procedure for outlier removal and rank

normalization, as well as their potential impact on the results.

Values > 5 s.d. values from the mean were deemed to be erroneous and were consequently removed before the main analysis (*Nat. Immunol.* 24, 408 – 411(2023).). Outlier removal was performed to reduce the impact of extreme values that could disproportionately influence the structure of the tree mapping, especially considering that the original Scottish tree model was developed from normalized and relatively homogenous clinical datasets. We followed a commonly accepted practice in large-scale clinical cohort analyses (*Nair.et al. & Nat Med* 31, 534–543 (2025).) by excluding individuals whose variable values exceeded 5 standard deviations from the cohort mean (computed within the Chinese cohort for each variable). This step helps reduce noise and ensures that the data used are within a clinically interpretable range. The number of excluded cases for each variable is now presented in Supplementary Figure 1. From a clinical perspective, these outlier values are often the result of recording errors, extreme comorbid conditions, or rare metabolic states and thus are not representative of the general T2D population.

This table shows the individuals with outlier data (total 575 individuals):

BMI	SBP	DBP	HbA1c	Cr	HDL-C	LDL-C	TRIG	ALT	Heart rate
3	6	6	0	627	14	12	553	716	23

Rank normalization was applied to each of the nine input variables (HbA1c, BMI, TC, HDL-C, ALT, creatinine, SBP, DBP, and TRIG) to align their marginal distributions with the Scottish tree population, which had undergone similar preprocessing. We also touched with Ewan Pearson about this question: In our pipeline, rank normalization was applied to each feature (row) across all samples (columns). Specifically, for each feature, we assigned ranks to the observed values across individuals, effectively transforming the feature's distribution to a uniform one on the rank scale (e.g., 1 to N, where N is the number of samples). Tied values were assigned the average of their ranks.

The main motivation for performing rank normalization is to ensure that every feature (row) has a non-zero standard deviation, which is critical for the downstream DDRTree algorithm (<https://doi.org/10.48550/arXiv.2407.01800>). DDRTree requires an initial PCA step, which cannot be performed if any row in the input matrix has zero variance. In our dataset, certain clinical variables exhibit very low variation across all patients, or their variation is nearly eliminated after regressing out covariates such as age and sex. Without normalization, this may result in rows with zero standard deviation, causing the PCA step in DDRTree to fail with an error such as “all rows have standard deviation zero.”

```
Code: rank_matrix = t(apply(feature_table, 1, rank))
```

```
rank_matrix = as.matrix(rank_matrix)
```

Here, each feature's raw values are replaced by their ranks (from 1 to N, where N is the number of samples), effectively “breaking ties” and spreading out values uniformly along the rank scale. This ensures that every row (feature) has a strictly positive standard deviation, which allows PCA to proceed without error. Rank normalization thus preserves the relative ordering of values while guaranteeing numerical stability for dimensionality reduction. Compared with directly using raw clinical values or residuals after regressing out age and sex—some of which can be constant or nearly so—rank normalization fundamentally prevents the zero-variance problem and is therefore indispensable for our model pipeline.

We acknowledge that these steps increase pipeline complexity. However, our goal was to minimize confounding effects arising from cross-population distributional differences, rather than to optimize model performance per se.

We have updated the **Methods section** to incorporate these clarifications and added the relevant sensitivity analyses to the supplementary material.

9. Methods: There are many examples of sentences where one asks multiple questions

regarding justification when reading it, one example being the sentence “the input features derived from a dataset consisting of 45,589 patients with T2D and 11,340 controls without diabetes, which was sourced from the CRDS database by random selection of 10% of the participants and exclusion of those who had important data missing”. What input are we talking about? Why are you considering controls and why this ratio? What is considered ‘important’ data to be missing?

Response: We thank the reviewer for this important comment and agree that the original sentence was unclear. We have revised the **Methods section** (Page 23) to clarify the definition of input features, the rationale for including non-diabetic controls, the selection ratio, and the criteria used for handling missing data.

In this context, the term “input features” refers to the clinical variables listed in Supplementary Table 2, which include anthropometric, metabolic, renal, hepatic, hematologic, comorbidity, and medication-related indicators. These variables were selected for their potential relevance to cardiometabolic heterogeneity and are used as input for model training and validation.

The combined dataset of 45,589 individuals with T2D (10% random sample from the CRDS) and 11,340 non-diabetic controls (sample from the Anhui Provincial Health Center) was constructed to support exploratory feature selection and distribution analysis. The inclusion of controls was intended to enhance the contrast between diabetic and non-diabetic phenotypes, allowing for better identification of diabetes-specific patterns, particularly during unsupervised learning and latent space evaluation.

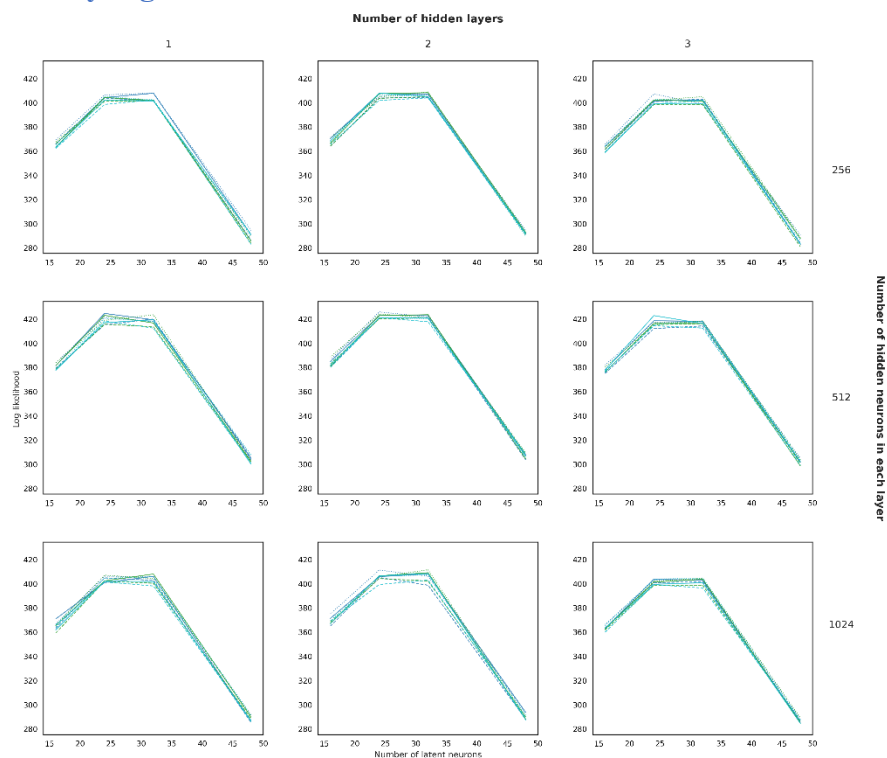
As for missing data, participants were excluded if they lacked any of the core variables required for clustering or mapping such as HbA1c, BMI, blood pressure, lipids, and creatinine, which we defined as “important” in the context of this analysis. This has now been clarified in the revised **Methods section** and noted in **Supplementary Table 2**.

10. Methods: A Variational Autoencoder (VAE) is introduced as a feature extractor. However, a justification as to why this is needed is lacking. While a VAE can be used as a feature extractor, it is a generative model partly trained to reconstruct its input. Its introduction as a feature extractor adds layers of complexity to the model pipeline such as: architecture design in terms of how many layers to include, regularization, latent dimensionality, training parameters (i.e. dropout, learning rate, batch size, etc.) – these details are also not given in detail nor justified in the manuscript. There are multiple machine learning methods perfectly capable of handling large feature inputs such as e.g. the LASSO or Elastic net. Investigations on whether simpler approaches would work are needed as the simplest possible method that fits the data well should be promoted.

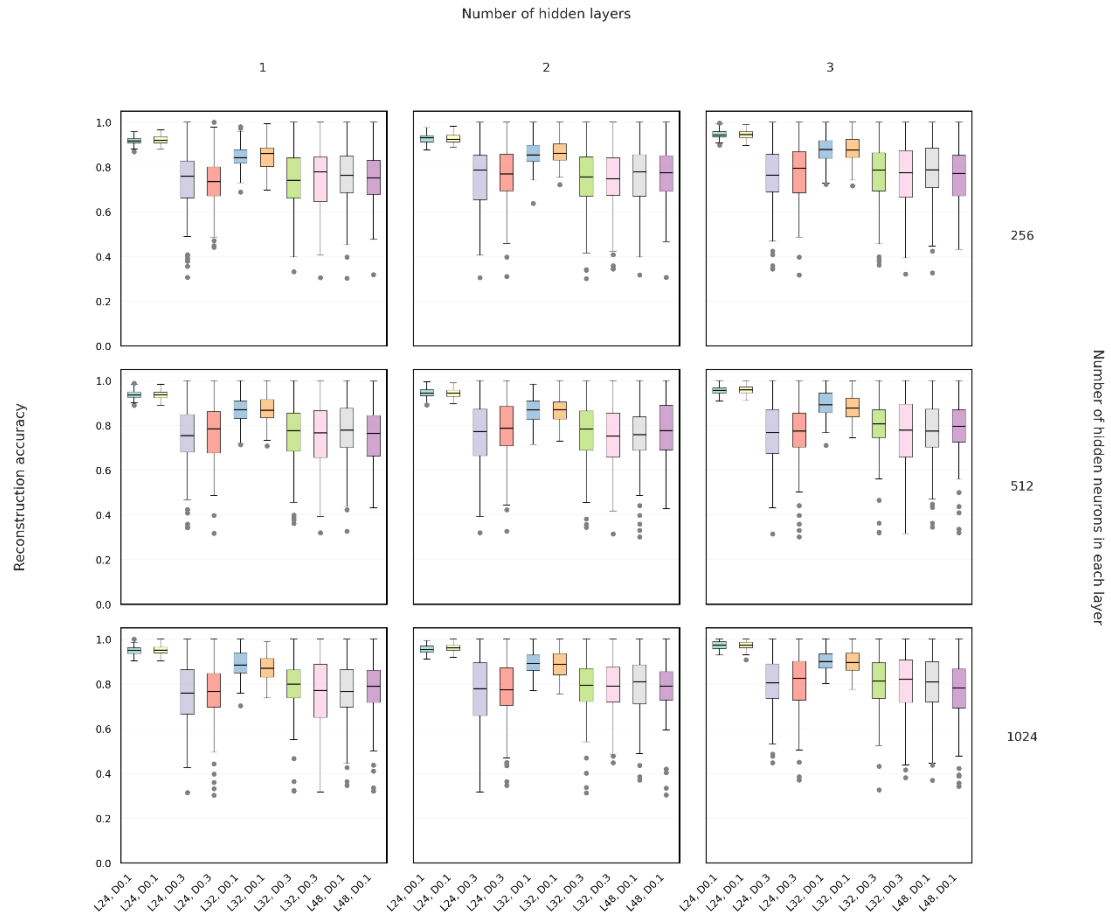
Response: We thank the reviewer for raising this important point. Our use of a variational autoencoder (VAE) as a feature extractor was motivated by the need to capture complex, non-linear relationships within high-dimensional clinical data in an unsupervised fashion. Unlike supervised approaches such as LASSO or Elastic Net, which are outcome-driven and optimize toward a specific label, our aim was to explore intrinsic data structure and reduce dimensionality without relying on predefined outcomes, which aligns with our goal of identifying phenotypic heterogeneity in T2D.

We acknowledge that VAEs are generative models and introduce architectural complexity. To address this, we have now included a detailed description of the model architecture, loss function, training procedure, and hyperparameter selection in the revised Methods section.

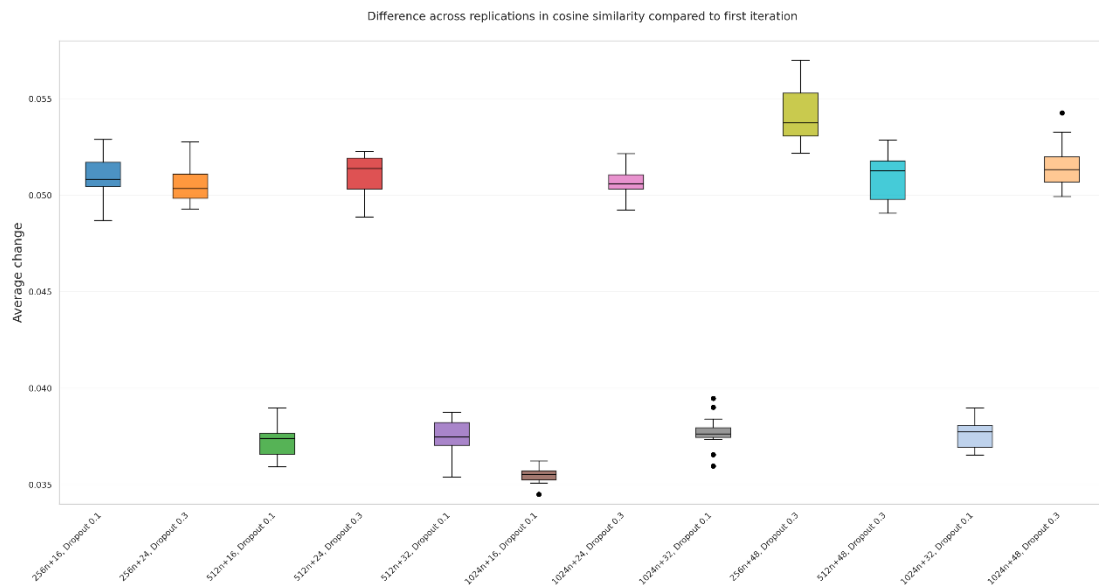
To address the concern of complexity, we conducted a systematic grid search to optimize model architecture and regularization, and presented the results in **Supplementary Figure 4**.



Supplementary Figure 4a illustrates the test-set log-likelihood across combinations of latent dimensions (15–50), hidden layer sizes (256/512/1024), number of layers (1/2/3), dropout rates (0.1/0.2), and KL-weight β ($1e-5$ to $1e-3$). The best generalization was observed in the configuration with one hidden layer of 512 neurons, dropout = 0.1, and $\beta = 1e-4$.



Supplementary Figure 4b shows the reconstruction accuracy on the training set, confirming that this configuration also offers the highest accuracy with the lowest variance.



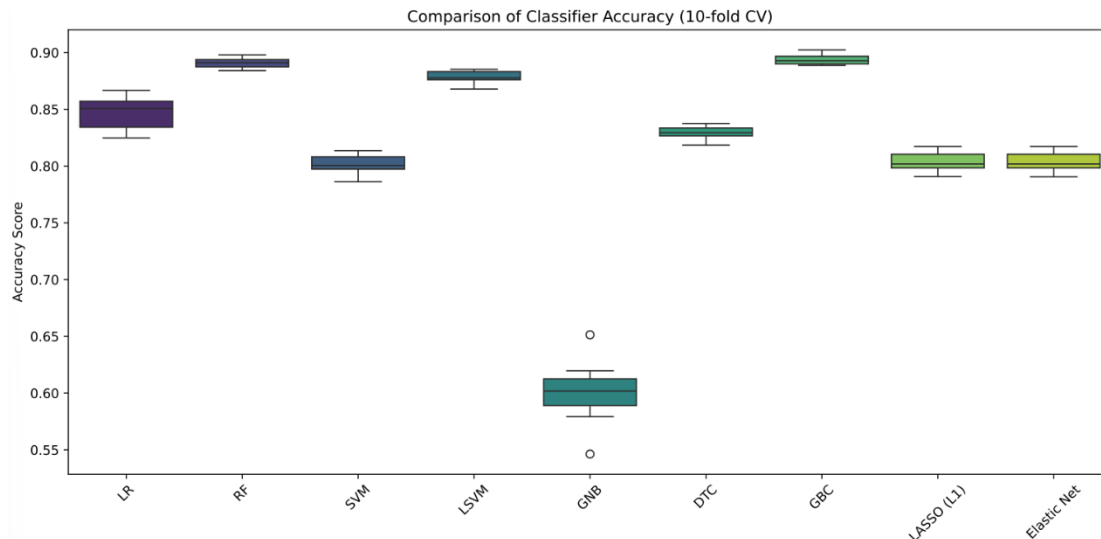
Supplementary Figure 4c evaluates the run-to-run stability across 5 training replications, with cosine similarity of the encoder outputs confirming minimal variation in the selected architecture.

These analyses were designed to ensure that the feature representations learned by the VAE were robust, generalizable, and stable. We have now added a summary of these results to the Methods section and clarified our rationale for using a VAE.

Importantly, the rationale for adopting VAE over simpler feature selection methods lies in its ability to capture nonlinear and potentially latent relationships in the data, which are often missed by traditional linear models. While the original tree model used nine clinician-selected variables, our goal was to employ a data-driven approach that could learn representations from complex interactions among features, especially in a large and heterogeneous population such as ours.

To address the reviewer's suggestion regarding simpler models, we also implemented LASSO and Elastic Net logistic regression models on the same dataset. As shown in **Supplementary Figure 5a**, the best ROC AUC and PR AUC achieved by these models were consistently below 0.75, significantly lower than those achieved using the VAE-

derived features with gradient boosting classifier (ROC AUC > 0.91, PR AUC > 0.96). This suggests that linear shrinkage models were insufficient to capture the underlying structure of the data for our task.



In summary, while we agree that simpler models should be considered first when appropriate, our extensive tuning, model comparison, and validation justify the use of VAE as a suitable and effective feature extractor in this context. We have revised the **Methods** section to include these additional details, and we added the relevant figures and tables to the supplementary material for full transparency.

11. Methods: There is a lack of detail in the description of the methods to a point where reproducing the experiments are impossible.

Response: We appreciate the reviewer's comment on the need for more detailed methodological descriptions to ensure reproducibility. In response, we have substantially expanded the Methods section by providing comprehensive details including:

Cohort selection and inclusion/exclusion criteria with exact ICD codes and demographic filters. Clinical variables used for analysis, with data preprocessing steps such as outlier removal (beyond 5 s.d. values), missing data handling, and rank normalization detailed.

VAE architecture and training protocol, including network layers, activation functions, dropout rates, latent space dimensionality, loss functions, optimizer settings, number of epochs, and batch size.

Feature extraction process from the trained VAE and downstream machine learning classifiers with hyperparameter settings and cross-validation procedures.

Software tools and packages used, with versions to ensure computational reproducibility. Statistical analysis methods and metrics reported, such as accuracy and classification reports.

Additionally, we have provided the complete model training and evaluation code to facilitate direct reproduction of our experiments. We have made considerable revisions to both the main manuscript and the supplementary materials. If there remain any sections that are unclear or require further clarification, we would be truly grateful for the opportunity to address them. We sincerely appreciate your thoughtful and constructive feedback.

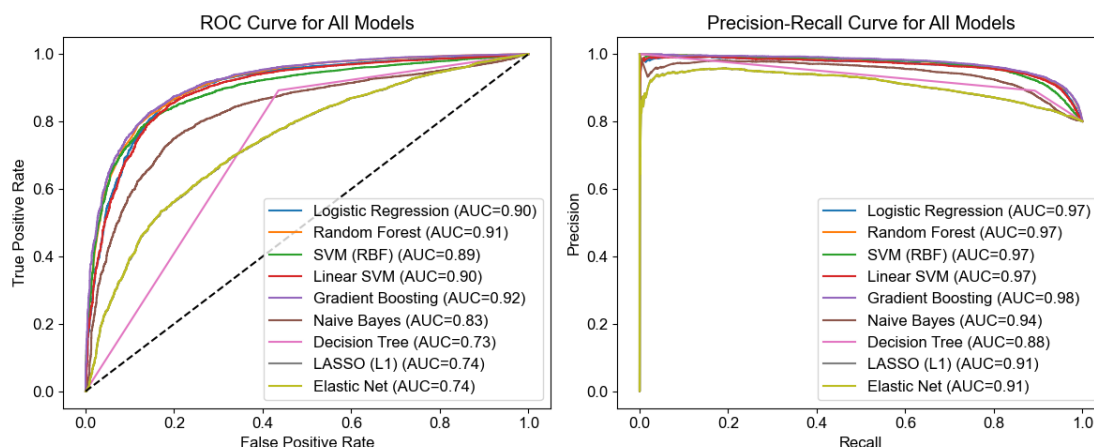
12. Methods: It is claimed that the GBC method achieves “highest accuracy” and therefore is chosen as a “final model”. However, the authors are considering a multitude of outcomes so is that really the case for ALL of these? In addition, is accuracy really the best measure to consider? One could imagine that some of the considered outcomes are rare leading to class imbalance. Alternative measures not so sensible to class imbalance (e.g. ROC or PR AUC) should at the very least be considered for model selection, and at the very least reported. Further – there is no mention of how the prediction threshold was decided.

Response: We thank the reviewer for this insightful comment. We acknowledge that accuracy alone may not fully capture model performance, especially given the potential class imbalance across multiple outcomes.

First, we would like to clarify that our study focuses on a binary classification task—

predicting diabetes status (T2D vs. non-T2D)—and not multiple distinct outcomes. Thus, model comparison and selection were based on this single binary endpoint.

Second, we acknowledge that relying solely on accuracy may be suboptimal in the presence of class imbalance. To ensure a fair and robust model comparison, we evaluated several classifiers using 10-fold cross-validation. In response to the reviewer’s suggestion, we have now evaluated and reported additional performance metrics, including the area under the ROC curve (ROC AUC) and precision-recall AUC (PR AUC), which are less sensitive to imbalance. These are now presented in the **Results section** and **Supplementary figure 5**. Furthermore, we have addressed the issue of threshold selection, we used the standard threshold. These additions strengthen the robustness of our model evaluation and align with best practices in binary classification under potential class imbalance.



We believe these additions provide a more robust and comprehensive assessment of model performance and selection criteria.

13. Methods: It is not specified if both the mean and variance of the approximate posterior gaussian are used as extracted features in the VAE pipeline, or simply the mean.

Response: In our VAE pipeline, the approximate posterior Gaussian distribution is parameterized by both mean (μ) and variance (log-variance) during training to enable variational inference and calculation of the Kullback–Leibler divergence loss term.

However, for downstream analyse, including latent space perturbation analysis and feature extraction—we use only the mean vector μ as the deterministic latent representation for each sample. This choice is consistent with common practice in VAE-based pipelines, where the mean vector serves as a stable embedding, avoiding stochastic variation introduced by sampling.

This approach is consistent with common practice in VAE-based feature extraction, where the mean vector serves as a deterministic embedding that effectively summarizes each sample’s latent features. The variance parameters primarily contribute to the regularization and uncertainty modeling during training but are not directly employed as features in subsequent analyses. We believe that using the mean latent vector alone sufficiently captures the essential features for our analyses.

We have updated the **Methods section** accordingly and thank the reviewer for highlighting this point.

14. Methods: The feature importance method is not adequately described.

We appreciate the reviewer’s comment regarding the clarity of the feature importance analysis. In the revised Methods section, we have now explicitly described the two complementary strategies used to estimate feature importance:

A weight-based approach, computing the sum of absolute encoder weights from each input to all hidden nodes;

A perturbation-based approach, adapted from SHAP analysis, that quantifies the impact of setting a feature to missing on the resulting latent representation.

These steps together allowed us to identify features that were both structurally influential in the encoder and practically impactful on the learned representation. We hope this clarification addresses the reviewer’s concern.

We have updated the **Methods section** accordingly and thank the reviewer for this point.

15. Code availability: Given the lack of detail in the description of the methods and no

code availability it is impossible to reproduce the presented results of the manuscript, even if one would have access to the data.

Response: We thank the reviewer for highlighting the importance of reproducibility. To address this, we have made the code for training the VAE and constructing the Chinese T2D subtype tree publicly available at: <https://github.com/Tongyue1999/Chinese-DDrtree>. The repository includes detailed scripts with all model parameters and training configurations to allow full replication of the results, assuming access to the CRDS data or an equivalent dataset.

Additionally, we have clarified this information in the revised ***Code Availability*** section of the manuscript.

Reviewer #2 (Remarks to the Author):

In this manuscript, the authors sought to characterize Chinese-specific T2D heterogeneity using deep learning-informed tree models.

The authors have performed novel analyses, compared their findings to a previously published Scottish study by Nair et al., and validated their findings in multiple sub-cohorts.

Their study provides a deeper understanding of the phenotypic variations in T2D in the Chinese population.

1. I think the unique risk factors for diabetes in the Chinese population should be mentioned in the introduction from the available literature. This will help clarify why this study is justified.

Response: We thank the reviewer for pointing this out. In response, we have now expanded the introduction to incorporate recent literature highlighting the unique clinical and pathophysiological characteristics of type 2 diabetes in Asian populations, including Chinese individuals, as follows.

It is accepted that Asian populations have unique physiopathological characteristics regarding T2D. (*Nat Rev Endocrinol.* 2022 Jul;18(7):413-432). Such characteristics reflect in different subtypes or patterns that emerge when efforts were made to reclassify T2D in Asian populations using methods establish in European descendents. For example, Xiong et al found two novel subgroups when clustering Chinese T2D patients, namely uric acid-related diabetes and inheritance-related diabetes (*J Diabetes Investig.* 2021 Aug;12(8):1346-1358.). Also, in Asian Indians, two new subtypes characterized in insulin resistance were identified. (*BMJ Open Diabetes Res Care.* 2020 Aug;8(1):e001506.) In our study, again, we discuss the lower BMI threshold for disease onset, reduced β -cell function, and higher propensity for visceral adiposity, which support the need for ancestry-specific classification models. Relevant references have been added to strengthen the rationale for developing a Chinese-specific T2D

classification model.

We have revised the **Introduction** accordingly in page 5 lines 113-115.

2. Although over 8.6 million individuals are in the cohort, only a part of it is used in the analysis. The sentence might be misleading to the readers.

Response: We thank the reviewer for pointing this out. We agree that the original sentence may be misleading. Although the multi-center cohort database includes over 8.6 million individuals, for the purpose of this study we focused on 32,501 individuals with newly diagnosed T2D who met the inclusion criteria. We have revised the sentence in the manuscript to clarify this.

3. In the introduction, 2nd paragraph, the second sentence, "we come," may be removed.

Response: Thank you for the suggestion. We have revised the sentence to improve clarity and flow. The phrase "we come to" has been removed as suggested.

4. I am not sure whether clustering is an arbitrary allocation. Even clustering methods utilize information from the input clinical parameters, while I do agree that discretization is of limited utility.

Response: We thank the reviewer for this insightful comment. We agree that clustering approaches, such as k-means or hierarchical clustering are indeed data-driven and utilize meaningful input features. Our intention was not to suggest that such methods are entirely arbitrary, but rather to emphasize that the process of forcing individuals into discrete and mutually exclusive clusters may not fully capture the continuous nature of disease heterogeneity. We have revised the text in the **Introduction** (Page 5 lines 104-107) and **Discussion** (Page 18 lines 441-450) to clarify this distinction and avoid misinterpretation.

5. In the CRDS cohort, the number of participants with long-term follow-up was relatively low. How might this have affected the tree structure and deep learning during training?

Response:

We appreciate the reviewer's question regarding the impact of limited long-term follow-up in the CRDS cohort. To clarify, the tree structure and deep learning model were trained exclusively on baseline (cross-sectional) data from newly diagnosed T2D patients. Specifically, we followed the methods described by *Nair et al.*¹¹ and mapped CRDS individuals onto the existing Scottish tree using nine baseline clinical variables (HbA1c, BMI, TC, HDL-C, ALT, TRIG, serum creatinine, SBP, and DBP). Deep learning-based feature selection based on characteristics (in the **Supplementary table 2**) and the construction of the Chinese-specific tree also relied solely on new ten baseline characteristics. Therefore, the number of participants with long-term follow-up does not affect the training or structure of the tree model or the feature selection process. However, the longitudinal data were used later to evaluate the phenotypic shifts and risk trajectories across branches in external validation cohorts, rather than during model development. We also revised the figure to clarify this question.

6. Also, please specify how many participants had completed 10-year follow-up data in the text when you first described it. I assume this proportion would be lower given that only 5.6% of T2D patients had 5 years of follow-up initially. However, I later see that the number of long-term follow-ups has increased, which introduces some confusion. It would be helpful for the authors to clarify this discrepancy.

Response: We thank the reviewer for this point and appreciate the opportunity to clarify. Specifically, a total of 317 individuals had completed the 10-year follow-up. The predicted 10-year probabilities of complications reported in the Results section (e.g., myocardial infarction, stroke, MASLD) are model-based estimates derived from time-to-event analyses using cause-specific hazard models and competing risk regression.

These models were constructed using available longitudinal data, primarily from patients with 1- to 5-year follow-up, and then extrapolated to estimate 10-year risks across the population of interest ($n \approx 26,000$) based on each individual's baseline characteristics and mapped position on the tree. This approach allows us to assess how the predicted risk of complications varies across the phenotypic landscape defined by the tree dimensions. The long-term follow-up data were used only to see the change in the phenotypic progression across the tree.

We have revised the manuscript to explicitly state that the predicted 10-year complication probabilities are model-derived estimates (Page 28-29 Section: Evaluating the distribution of phenotypes and outcomes).

6. Additionally, how might this variation in follow-up duration affect the algorithm's ability to learn patterns effectively, particularly for predicting long-term outcomes such as diabetic retinopathy?

Response: Thank you for this important question. We would like to clarify that our tree-based model and deep learning feature selection were developed solely using cross-sectional baseline data and were not trained to directly predict long-term outcomes such as DR. Therefore, variation in follow-up duration does not affect the algorithm's ability to learn phenotypic patterns or construct the tree structure.

Long-term outcomes, including DR, were assessed post hoc using time-to-event data through competing risk regression, stratified by position on the tree. Although follow-up duration varied among individuals, such censoring is appropriately handled within the survival analysis framework, allowing us to estimate cumulative incidence and predicted risk over 10 years. The variation in follow-up time may affect the precision of certain risk estimates, particularly in later time intervals, but does not compromise the integrity of the phenotypic representation or the model's ability to stratify risk based on baseline clinical features.

We have added this clarification to the manuscript to avoid confusion.

7. The observation was that HbA1c had modest effect, while lipids had the most substantial effect. This is surprising, as one would expect HbA1c to be the strongest driver when studying diabetes outcomes.

Response: We thank the reviewer for this thoughtful comment. Indeed, it is commonly assumed that HbA1c would be a dominant factor in shaping diabetes phenotypes due to its central role in glycemic control. Our findings and those reported in the original Scottish study published in *Nature Medicine* (*Nair et al., 2022*)—consistently show that HbA1c has relatively modest influence on the tree structure compared to other metabolic variables such as triglycerides, HDL cholesterol, or liver enzymes.

We believe this reflects an important conceptual distinction between differentiating diabetes from non-diabetes (where HbA1c is key) versus stratifying heterogeneity within individuals already diagnosed with T2D, which was the focus of our study. At the time of diagnosis, most individuals meet similar glycemic thresholds (HbA1c >6.5%), resulting in relatively constrained variance in HbA1c values. In contrast, lipid markers such as triglycerides and HDL cholesterol, as well as variables related to blood pressure and liver function, display greater interindividual variability and are more informative for capturing metabolic phenotypes and predicting future complications.

Additionally, HbA1c reflects short-to-intermediate term glycemic status, but may not fully represent underlying metabolic dysfunction or risk trajectory, especially in populations where insulin resistance and dyslipidemia play a predominant role in disease progression. This pattern is also consistent with prior literature showing that lipid abnormalities and blood pressure are major drivers of macrovascular and hepatic complications in T2D, sometimes even more so than glycemia alone.

We have added this explanation to the **Discussion** section (Page 17) to clarify this point for readers.

8. Also, liver disease exerts an undue influence on specific phenotypes such as DR and CVD. I think this is an artifact arising from the huge burden of liver disease in the

Chinese population. However, I am not denying that it could indeed have a significant influence on diabetes and its outcomes. Therefore, the authors must consider reporting the burden of liver diseases in the sample for each diabetes complication explored in this manuscript as a table.

Response: We thank the reviewer for the valuable suggestion regarding the influence of liver disease on diabetes complications. We acknowledge the high burden of liver disease in the Chinese population, which may contribute to the observed associations with certain phenotypes such as DR and CVD.

To address this, we have added **Supplementary table 5** summarizing the prevalence of MASLD and cirrhosis among patients with various diabetes complications in our cohort. This table demonstrates the significant co-occurrence of liver disease and diabetes complications and provides important context for interpreting our phenotypic clustering results.

Diabetes Complication	Total Events (n)	MASLD Cases	Cirrhosis Cases
Myocardial Infarction	213	69	3
Stroke	2,187	754	36
Ischemic Stroke	1,984	661	34
Intracerebral Hemorrhage	139	45	4
Heart Failure	1,333	413	33
Heart Failure with Reduced EF (HFrEF)	97	23	1
Heart Failure with Preserved EF (HFpEF)	344	98	7
Diabetic Retinopathy	1,856	546	29
Chronic Kidney Disease	1,222	327	38
End-Stage Renal Disease	216	49	11
Diabetic Peripheral Vascular Disease	655	546	13

We have also expanded the **Discussion section** to include consideration of these population-specific comorbidity patterns and their implications for precision diabetes management.

9. BMI differences between ethnicities could introduce specific errors, especially in

cases like this, where there is evidence of interplay of BMI with other risk factors. Does this also imply that the effect of BMI on diabetes phenotypes may have been masked due to the mismatch?

Response: Thank you for this insightful comment. Indeed, your observation aligns closely with our previous findings. When we mapped the Chinese patients onto the Scottish tree model, we made extensive efforts to match the distribution of baseline clinical variables as closely as possible. However, we still observed a notable absence of patients in the upper right region of the tree, which in the original Scottish model (*Nair et al., Nat Med 2022*) corresponds to individuals with very high BMI.

This gap clearly indicates a mismatch in the BMI distribution between the Chinese and Scottish populations. Given that BMI interacts with other metabolic risk factors and significantly shapes diabetes phenotypes, such mismatch likely masks or alters the apparent effect of BMI on T2D subtypes when using a European-derived model in a Chinese cohort.

This discrepancy is precisely why we emphasized the need to build a Chinese-specific tree model that better reflects the unique anthropometric and metabolic characteristics of the Chinese population. The generally lower BMI among Chinese patients compared to Europeans means that phenotypic clusters, and their associated risk profiles, which differ substantially. Without accounting for these differences, cross-ethnic application of classification tools risks misclassification and reduced clinical utility.

Therefore, our study not only demonstrates the limitations of applying European-derived models to non-European populations but also highlights the importance of developing ancestry-specific tools to capture true underlying heterogeneity and improve precision diabetes care.

We have also expanded the **Discussion section** (Page 17).

10. The authors have not stated explicitly in the text, or supplementary which variables were considered for selecting important features.

Response: Thank you for the valuable comment. The variables considered for feature selection encompassed anthropometric measures, glycemic and lipid markers, renal and liver function indices, hematologic parameters, comorbidities, and medication use. To improve transparency and reproducibility, we have now provided a comprehensive list of all variables included for feature selection in the revised **Methods section** (Page 23, lines 583-585). Additionally, these variables are summarized in Supplementary Table 2.

11. In the following line, BMI was repeated in the text.

Response: Thank you for your careful review. We have identified the error where "BMI" was mistakenly repeated in the sentence and have corrected it in the **Results section** accordingly.

12. Predicting the long-term risk of diabetic retinopathy (DR) remains challenging when relying solely on traditional risk factors (Figure 1). A higher risk for diabetic retinopathy (DR) was observed in individuals with high HDL-C and indicators of liver function. However, these findings are not well-supported by existing large-sample studies in the Chinese population. Furthermore, high HDL-C levels are generally considered protective against diabetic retinopathy (DR) due to their anti-inflammatory properties. However, conflicting findings—primarily from cross-sectional studies—do exist. The authors should acknowledge these discrepancies and offer a possible explanation for the observed association in the discussion, supported by appropriate references. Genetics alone cannot account for such heterogeneity, particularly when the phenotype associations come into question.

Response: We thank the reviewer for this insightful comment. We agree that the observed association between higher HDL-C levels and increased DR risk appears to contradict the traditional understanding of HDL-C as protective due to its anti-inflammatory effects. In response, we have added a paragraph in the Discussion to acknowledge this discrepancy and provide potential explanations, citing relevant literature.

We know that the association between HDL-C levels and diabetic retinopathy (DR)

remains inconclusive across studies. While the DCCT/EDIC study reported an inverse relationship between HDL-C and DR severity in type 1 diabetes (*Ophthalmol. Vis. Sci.* 2004;45:910–918), a cross-sectional study by Sasso et al. found that higher HDL-C levels were independently associated with increased DR risk in type 2 diabetes(*Diabetes Res. Clin. Pract.* 2019;150:236–244.). In contrast, other large-scale studies, including a multinational case-control study and population-based cohorts such as the Singapore Malay Eye Study, found no significant association between HDL-C and DR after adjusting for confounders (*Circulation.* 2013;129:999–1008. & *Ophthalmology.* 2008;115:1869–1875.).

We also highlight that the co-existence of liver dysfunction, known to affect HDL composition and function, may play a role in mediating this paradoxical relationship. The revised **Discussion section** (Pages 15-16, lines 373-388) now includes these interpretations and references to support them.

13. What outcomes were different between CRDS and JADE? Please add to the main text.

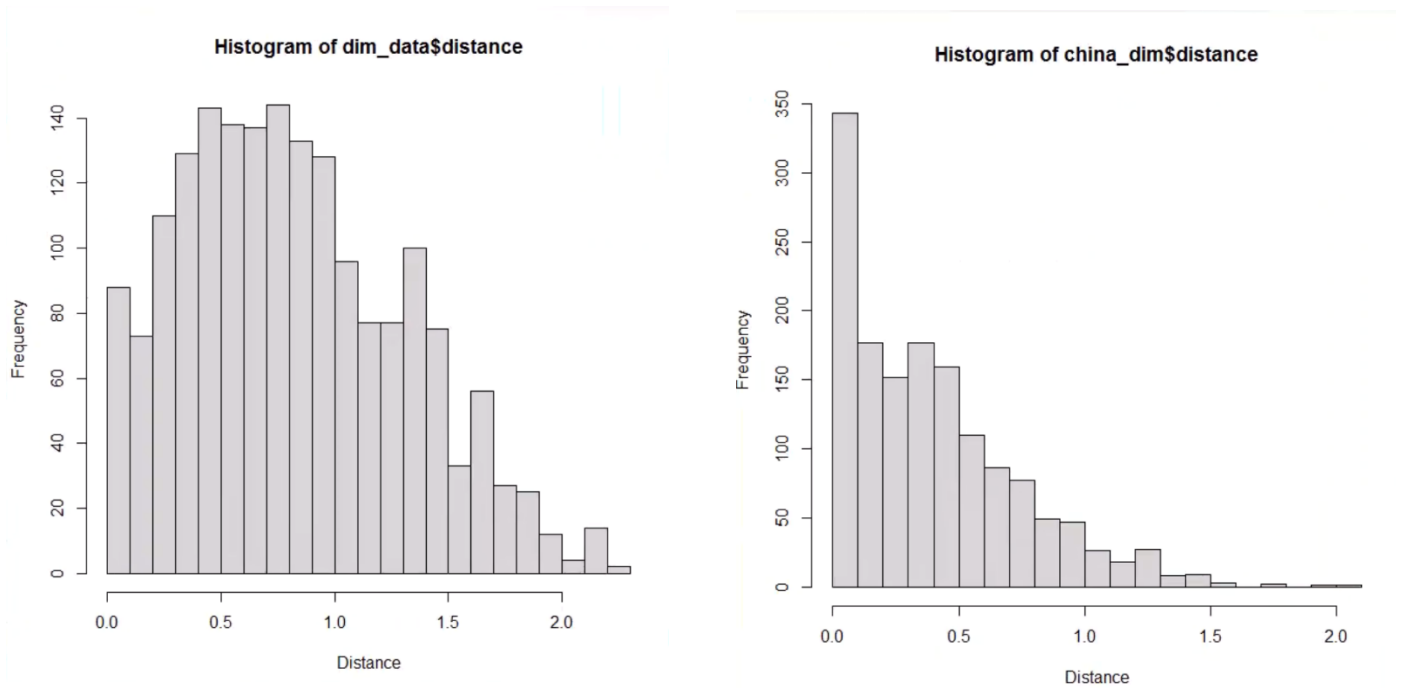
Response: Thank you for your valuable comment. We have now added a detailed comparison of key clinical outcome distributions between the CRDS and JADE cohorts in the Results section (see paragraph **Validation of the Chinese Tree**). Briefly, while heart failure (HF) and myocardial infarction (MI) showed similar distribution patterns in both cohorts, stroke and ischemic stroke (Istroke) patterns differed. Heart failure with preserved ejection fraction (HFpEF) was similar to stroke in the CRDS cohort, whereas in the JADE cohort, HF and stroke shared similar patterns, and MI and Istroke clustered similarly. Additionally, patients with cirrhosis were mainly concentrated in branches 1 and 3 in both cohorts, and both end-stage renal disease (ESRD) and chronic kidney disease (CKD) were enriched in branches 5 and 6. We have included these findings in the **Results** to better illustrate cohort differences and the robustness of our model across diverse healthcare systems.

14. More Scottish participants shifted their position after 5 years compared to the

Chinese participants. Again, we would expect that the Asian population is likely to be more vulnerable. What could be the reason behind this finding?

Response: Thank you for this important observation. We apologize for the confusion and would like to clarify that the comparison was based on the same set of Chinese participants projected onto two different tree models: one derived from the Chinese cohort and the other from the Scottish cohort. Interestingly, a larger proportion of Chinese individuals appeared to shift their positions more substantially on the Scottish tree than on the Chinese tree after five years.

Several factors may account for this question. First, the Scottish tree, being derived from a different population with distinct metabolic profiles and progression dynamics, may represent a broader latent space with greater variation in trajectories. When the Chinese participants were projected onto this space, their follow-up positions might have appeared to change more because of differences in the structure and spacing of the latent dimensions. Second, regions of the Scottish tree, such as those corresponding to individuals with high BMI and poor glycemic control, which may have had sparse or no counterparts in the Chinese baseline data, leading to initial clustering near the center and greater observable displacement at follow-up. Third, this finding underscores the impact of population-specific characteristics on the learned representation. Even when the same individuals are mapped onto different latent structures, the tree's topology and feature scaling can influence the apparent magnitude of change. Lastly, when comparing the distributions of distance (i.e., the Euclidean displacement in latent space), we found distinct patterns between the two cohorts. As shown in the boxplots, individuals in Scottish tree demonstrated a wider spread in movement (ranging from 0 to >2), whereas Chinese tree exhibited a more concentrated distribution.



To clarify this point, we have added an explanation to the **Discussion** sections (Pages 17, lines 423-429).

15. Was any data curation performed for the phenotypes? None has been mentioned except outlier identification and corresponding ICD codes.

Response: Thank you for your important question. Our data preprocessing followed procedures consistent with those described by Nair et al. (2022). Specifically, we performed outlier removal to exclude extreme values likely due to measurement or recording errors. We also curated the cohort by selecting individuals and defining diagnostic outcomes based on standardized ICD codes to ensure phenotypic accuracy. In addition, certain diseases were excluded during the selection process to avoid confounding; the corresponding ICD codes for these exclusion criteria are listed in **Supplementary Table 1**. Furthermore, during the DDRTree dimensionality reduction process, all features were normalized to harmonize scales across variables and reduce bias in downstream analyses. These steps together constitute a comprehensive data curation pipeline that ensures the reliability and consistency of the phenotypes used in our study.

We have added these in the **Method section** and **figure1** to better address this important point.

16. What retinopathy grades were included in the analysis (e.g., proliferative DR, any DR, etc.)? The phenotype associations are likely to change as per the DR definitions.

Response: Thank you for this important point. In our Chinese cohort, DR diagnoses were obtained from the hospital-based electronic health record (EHR) system, rather than through a centralized screening program or GP-based coding system. Consequently, DR was recorded as a single diagnostic entity without further grading or subtyping (e.g., non-proliferative vs. proliferative DR). We therefore treated DR as a composite outcome, encompassing all forms of diabetic retinopathy identified in the clinical setting. However, it is worth noting that in the Chinese healthcare situation, DR diagnoses captured in hospital EMRs are typically made during ophthalmology visits and are more likely to reflect moderate to severe stages, such as vision-threatening DR (VTDR) or proliferative DR (PDR), rather than early non-proliferative changes. This may partly explain why the 10-year cumulative incidence of DR in our study (5.7%) appears lower than the rates reported in population-based studies (often >15%) that include all grades of DR (*Nature Communications*. (2023) 14:4296). We fully acknowledge that the absence of DR grading may introduce heterogeneity in outcome ascertainment, and agree that phenotype associations could vary depending on the definition and severity of DR. Due to the limitations of the EHR structure and data availability, we were unable to stratify DR by severity in this analysis. We have added this point as a **limitation** (Pages 17, lines 367-37 and Pages 19, lines 483-486) in **Discussion** part of the revised manuscript.

17. The VAE model was trained on a dataset in which the number of controls was four times smaller than the number of cases. How might this imbalance affect the model's learning accuracy? So, does this mean the data was trained on more than 56,000 individuals? Is that number mentioned in the text, and is it consistent throughout?

Response: Thank you for your thoughtful questions. The VAE model was trained using

data from 45,589 individuals with T2D and 11,340 normoglycemic individuals (control). The control group was sourced from the Health Management Center of Anhui Provincial Hospital. The imbalance between case and control groups in our dataset reflects the real-world distribution of the condition under investigation, and we deliberately avoided artificial undersampling or oversampling in order to preserve the natural data characteristics. We acknowledge that such imbalance may affect the performance of supervised models if used in isolation. However, in our approach, we first applied a VAE to perform unsupervised feature extraction without regard to outcome labels, followed by a supervised Gradient Boosting Classifier (GBC) to identify features associated with T2D. This combination helps mitigate the influence of class imbalance by learning a more representative latent structure before classification. Regarding the sample size, and this figure 1 has been clearly stated and consistently used throughout the manuscript. We have reviewed the text to ensure that the reported numbers are consistent across all sections.

18. I am looking at Figure 1 and would have expected a higher number of DR cases—only 8.6% after 5 years seems surprisingly low. Normally, one would see more than 20%. The dimension 2 is associated with obesity and lipids. That’s interesting!

Response: Thank you for your insightful comments. Thank you for your valuable comment. Thank you for your insightful comment. We appreciate your observation. In our dataset, the number of DR events was 1,856 among 32,501 individuals, which corresponds to approximately 5.7%. However, when focusing on more vision-threatening forms, including diabetic macular edema (DME; $n = 431$) and vision-threatening diabetic retinopathy (VTDR; $n = 1,673$), the total number of events increases to 2,104 across 50,564 individuals (approximately 4.1%) in T2D populations with a mean disease duration of 8.4 years (*Nature Communications* | (2023) 14:4296).

We agree that the lower overall DR incidence may reflect the relatively early and differences in diagnostic coding and DR subtyping. We have clarified this in the revised Methods section, and thank you for bringing it to our attention.

We think several factors may help explain this discrepancy:

1. In our study, diabetic retinopathy (DR) was identified based on ICD-10 codes E11.371, and E14.371 as documented in the hospital-based EHR system. These codes primarily reflect moderate to severe forms of DR with clinical significance, including moderate NPDR with diabetic macular edema (DME), severe NPDR, and PDR. Therefore, the DR cases captured in our analysis largely represent VTDR, rather than mild or background DR, which are often under-recorded or not coded in the EHR. Consequently, cases of early or mild non-proliferative DR managed in outpatient or community settings may have been underrepresented.
2. The dataset used for DR identification was based on diagnosis codes from inpatient and specialist outpatient records. Since patients with mild DR might not seek tertiary care or may be managed within primary care without being coded in hospital records, under-ascertainment is likely.
3. Several studies suggest that DR prevalence may differ between populations due to variations in screening coverage, glycemic control, and healthcare access. For example, a large population-based study in China reported relatively lower DR prevalence (16.3%, *Nature Communications* | (2023) 14:4296).
4. The reliance on administrative codes may lead to an underestimation of DR incidence. In contrast, studies reporting higher incidence often include active fundus screening data, which is more sensitive for detecting early or asymptomatic DR.

We have now clarified this in the revised **Methods** and **Discussion** sections.

19. There are numerous figures and frequently changing numbers throughout the manuscript, which makes it challenging for the reader to follow the results.

The authors may consider consolidating them to improve clarity and interpretability.

Response: Thank you for this helpful suggestion. In response, we have carefully reviewed all figures, tables, and numerical results throughout the manuscript. We have consolidated and simplified several visualizations to reduce redundancy and improve

clarity. In addition, we have revised the layout and presentation of key results to enhance readability and allow for easier interpretation by both reviewers and future readers. We believe these changes significantly improve the overall flow and accessibility of the manuscript. You will notice that we have made considerable revisions to both the main manuscript and the supplementary materials. If there remain any sections that are unclear or require further clarification, we would be truly grateful for the opportunity to address them. We sincerely appreciate your thoughtful and constructive feedback.

Reviewer #3 (Remarks to the Author):

In their manuscript, the authors used a novel two-step approach—first validating a previously published ‘Scottish tree’ model in a Chinese population, then constructing a Chinese-specific phenotyping tree using a variational autoencoder (VAE)-based feature selection framework.

This manuscript presents a comprehensive and innovative approach to characterizing the heterogeneity of T2D in a large Chinese population using a deep learning-based phenotyping framework.

By first validating the applicability of a previously published ‘Scottish tree’ model and then constructing a Chinese-specific tree based on VAE-based feature selection, the authors provide compelling evidence for ancestry-specific disease patterns of diabetes and complication risks.

The study leverages a massive electronic health record dataset, robust statistical models, and external validations across independent cohorts, significantly enhancing the credibility and generalizability of the findings.

Major comments:

1. ‘The data included all the patients whose data is accessible from the participating centers’ medical record systems. Were all the data during the period accessible in those centers? If not, what are the percentages of accessible data, and what determined the accessibility? In addition, it is good to know the distribution of these 24 medical centers.

Response: Thank you for raising this important point regarding data accessibility and center distribution.

The data in our study include all patients whose records were accessible from the participating 24 medical centers’ electronic health record (EHR) systems during the study period. While the centers aimed to capture all available data, accessibility

depended on the completeness and integration of EHR systems at each center, as well as patient visit patterns.

To clarify, the overall scale of the 24 centers is as follows:

- Number of participating hospitals after data cleaning: 24
- Number of inpatient patients: approximately 8.63 million
- Total inpatient visits: approximately 14.53 million
- Total volume of medical records: approximately 3.63 billion entries
- Patients with at least 1-year follow-up: 2.55 million
- Patients with at least 3-year follow-up: 1.30 million

Regarding diabetes patients specifically:

- Total diabetes patients identified: approximately 1.01 million
- Number of inpatient records for diabetes patients: 2,067,455
- Patients with at least one inpatient admission: 952,497
- Patients with at least two inpatient admissions: 375,067
- Patients with three or more inpatient admissions: 192,693

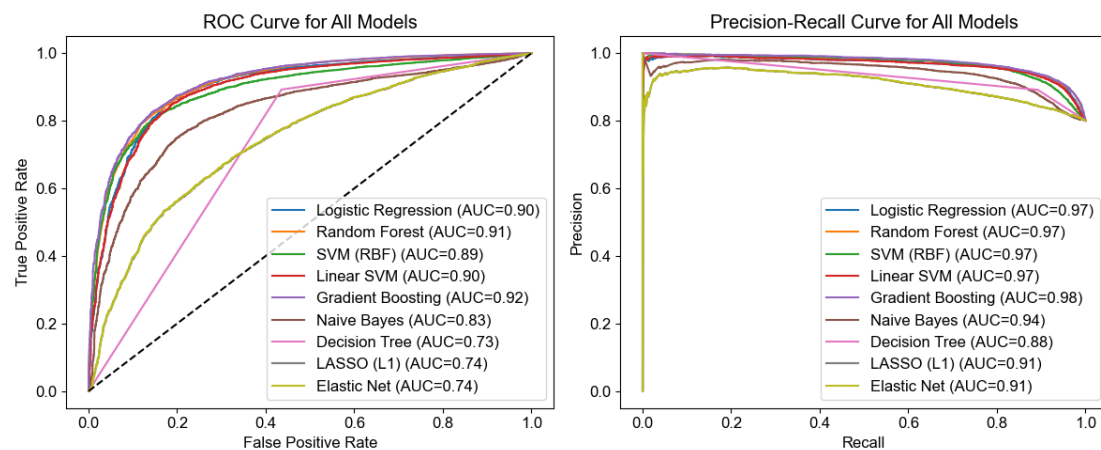
2. The rationale for selecting exactly ten features in the final model is not clearly explained. Why was this number chosen? It would be helpful if the authors could clarify whether they tested different numbers of input features and how this affected the tree structure, complication risk gradients, or model performance.

Response: We appreciate the reviewer's insightful question regarding the rationale behind selecting exactly ten features for the final model.

To identify the most relevant clinical features associated with T2D, we developed a feature selection framework that integrates a variational autoencoder (VAE) with machine learning. Using data from 45,589 individuals with T2D and 11,340 healthy

controls, the VAE model was trained on 48 clinical variables to perform dimensionality reduction. The model showed high reconstruction accuracy (median 0.85–1.0) and strong stability across training iterations. Feature importance was assessed based on encoder weights and confirmed by SHAP analysis. Both approaches consistently highlighted top features related to cardiovascular, hepatic, renal, and metabolic functions. The 10 most influential overlapping features identified were HDL-C, triglycerides, SBP, ALT, HbA1c, LDL-C, creatinine, heart rate, DBP, and BMI.

To validate the biological and predictive relevance of the ten features jointly selected by VAE-based weighting and SHAP analysis, we trained multiple machine learning classifiers to distinguish individuals with and without T2D. We compared the performance of several machine learning classifiers (including Logistic Regression, Random Forest, SVM, Gradient Boosting Classifier [GBC], LASSO and Elastic Net) based on accuracy, ROC AUC, and PR AUC metrics. Among them, GBC consistently achieved the highest classification performance (with Random Forest ranking second), and was therefore selected as the final classifier for feature importance evaluation.



Subsequently, we ranked all input features according to their importance weights in the trained GBC model and evaluated the model's performance using subsets of the top-ranked features. Specifically, we constructed feature sets ranging from the top 5 to the top 10 features:

```
feature_sets = {
    "5": ["HDL-C", "TG", "SBP", "ALT", "HbA1c"],
```

"6": ["HDL-C", "TG", "SBP", "ALT", "HbA1c", "LDL-C"],

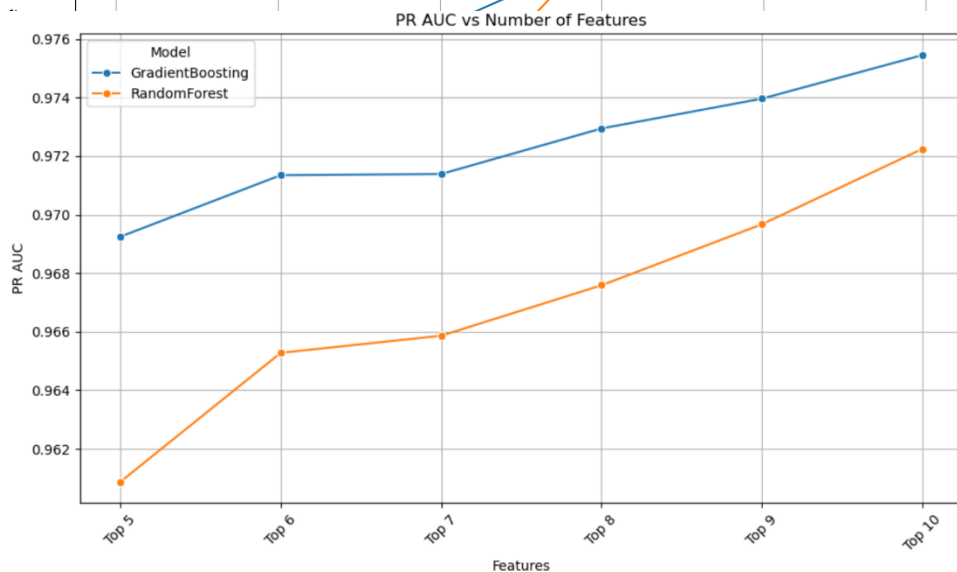
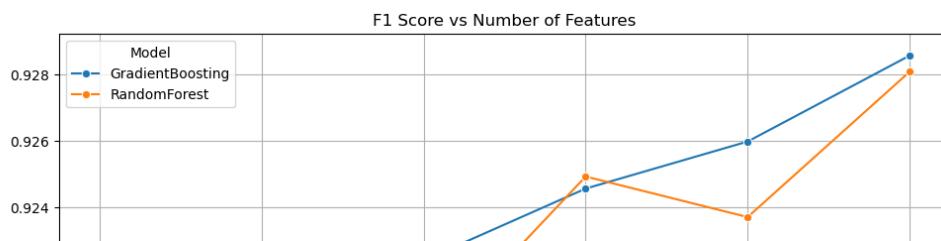
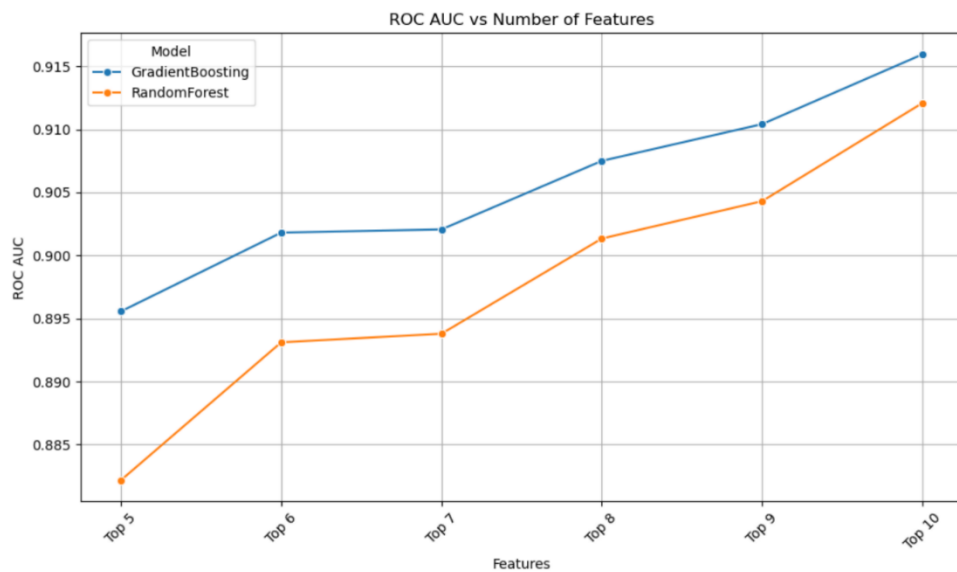
"7": ["HDL-C", "TG", "SBP", "ALT", "HbA1c", "LDL-C", "HR"],

"8": ["HDL-C", "TG", "SBP", "ALT", "HbA1c", "LDL-C", "HR", "BMI"],

"9": ["HDL-C", "TG", "SBP", "ALT", "HbA1c", "LDL-C", "HR", "BMI", "DBP"],

"10": ["HDL-C", "TG", "SBP", "ALT", "HbA1c", "LDL-C", "HR", "BMI", "DBP",

"Creatinine"]}

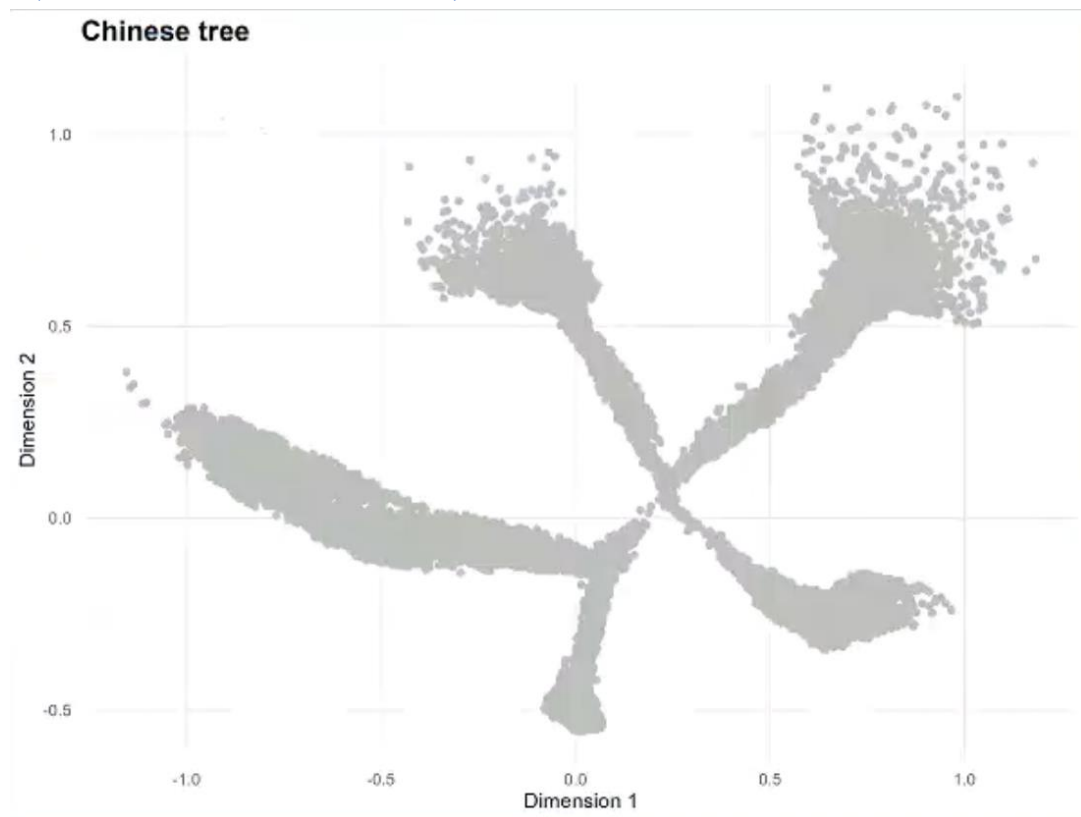


We assessed the classification performance of each subset using ROC AUC, PR AUC, and F1 Score. The results showed a gradual increase in performance as more top-ranked features were included, with the 10 subset yielding performance metrics very close to that obtained when using all available features. Thus, we determined that the top 10 features sufficiently captured the discriminative signals for T2D classification in the Chinese population, providing a balance between model simplicity and predictive accuracy.

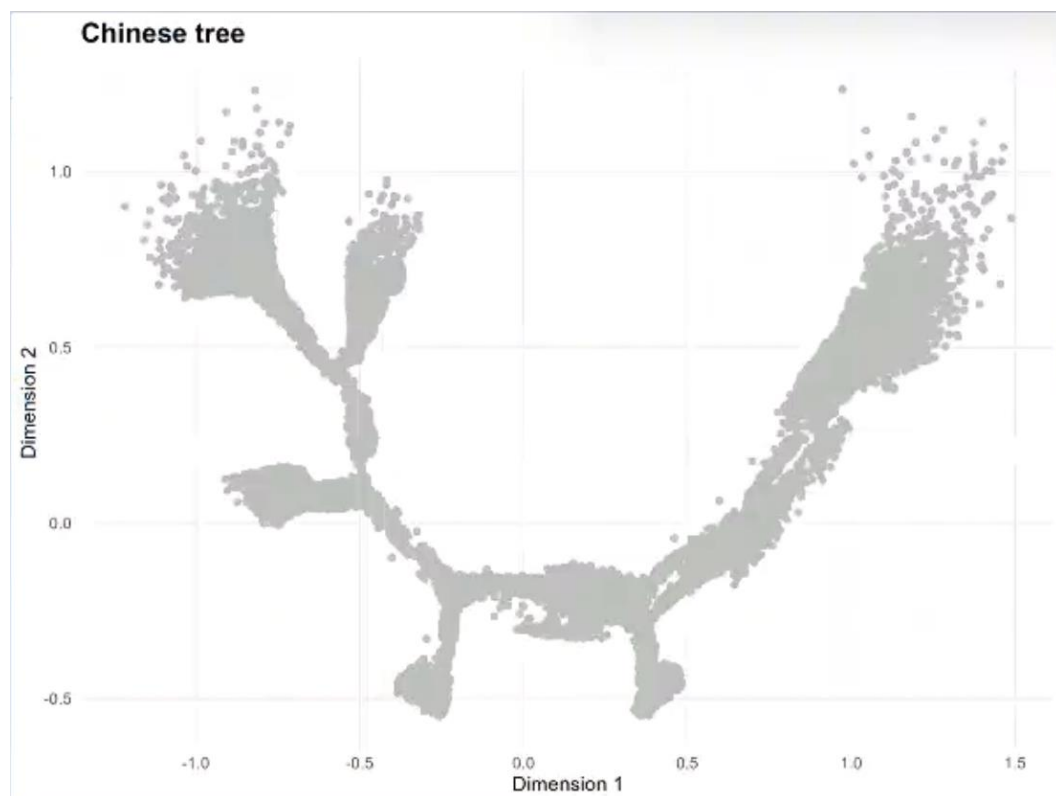
In response to the reviewer's concern about the impact on the tree structure and risk gradients: we indeed performed DDRTree visualization using different feature subsets from 5 to 9. The results demonstrated that the overall shape and branching structure of the tree were affected by the specific input features used. The 10-feature tree (our final choice) displayed the most stable and informative separation of patient subgroups, and also led to more distinct gradients in complication risks when used as input for subsequent Fine–Gray competing risk models.

And we show that tree constructed by features:

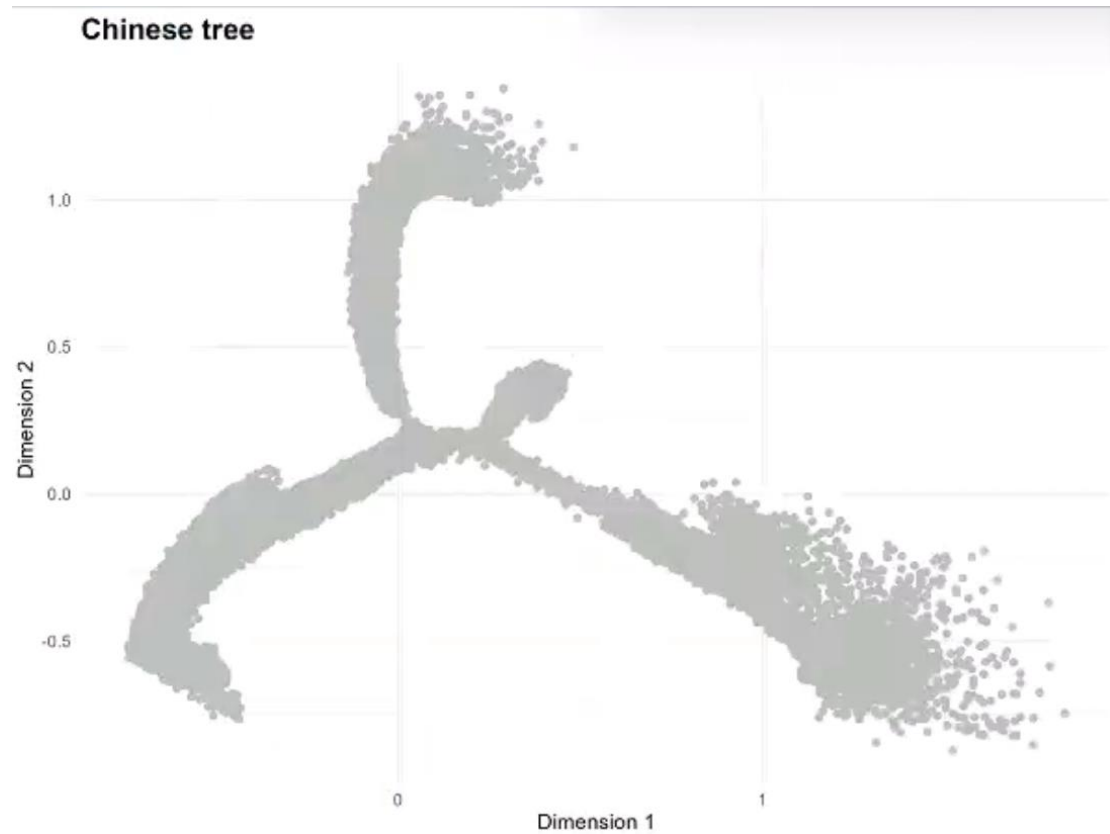
5 (HDL-C; TG; SBP; ALT; HbA1c)



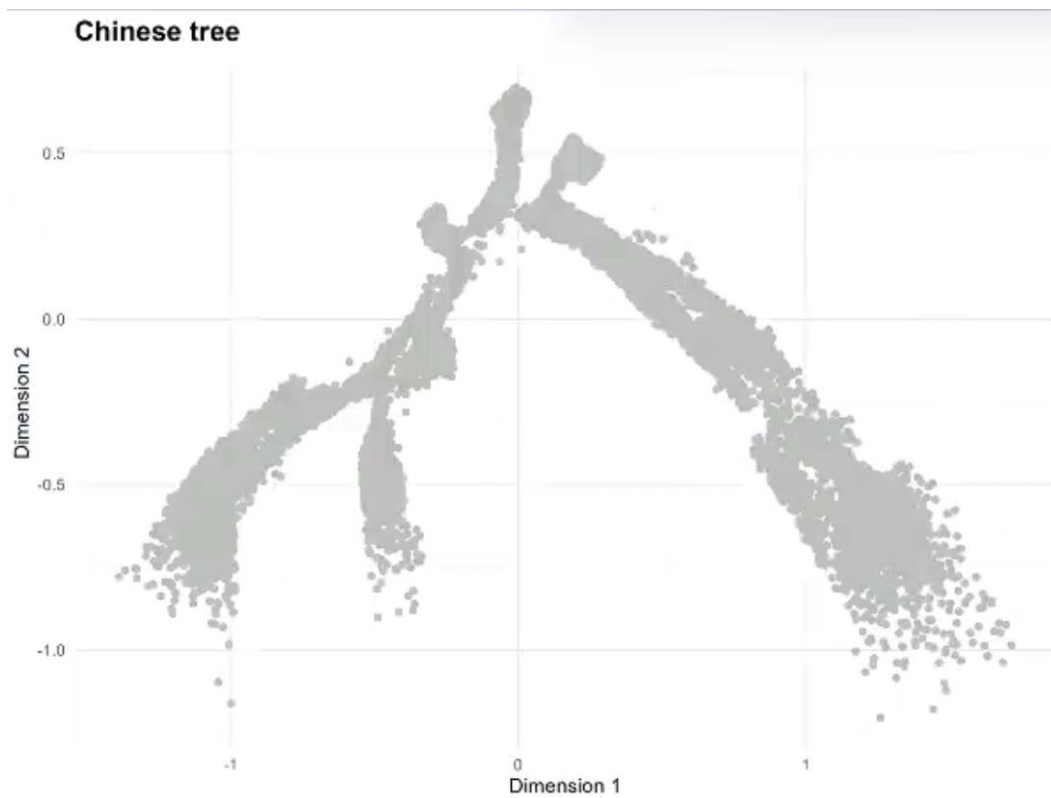
6 (HDL-C; TG; SBP; ALT; HbA1c; LDL-C)



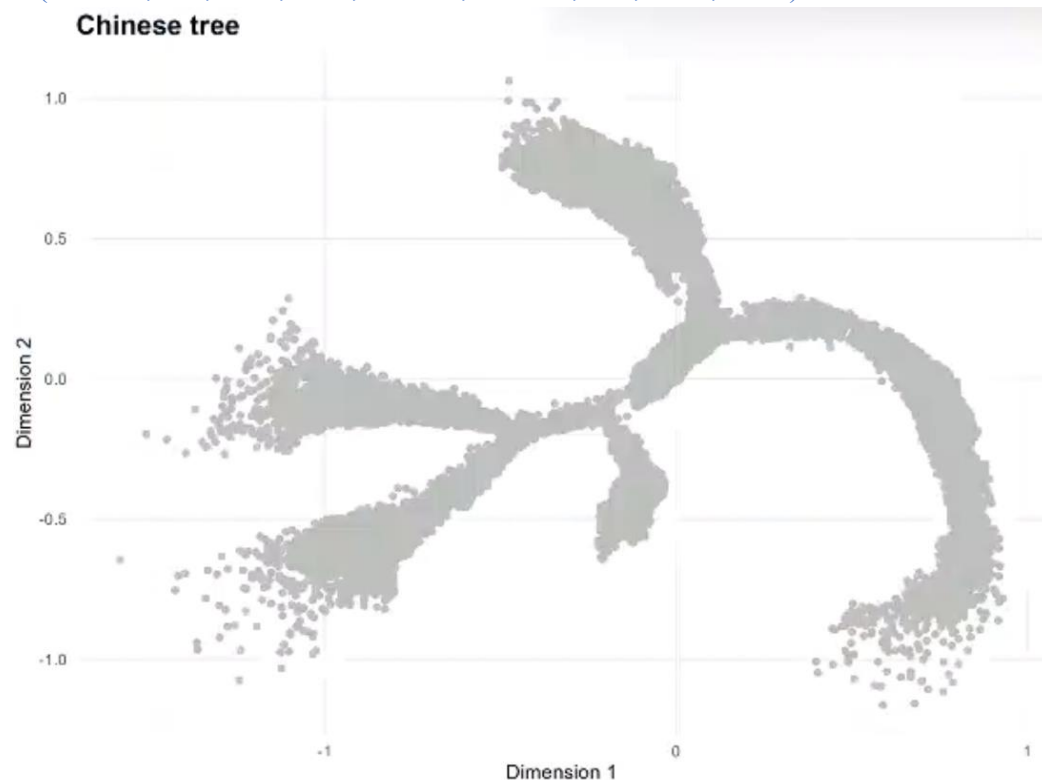
7 (HDL-C; TG; SBP; ALT; HbA1c; LDL-C; HR)



8 (HDL-C; TG; SBP; ALT; HbA1c; LDL-C; HR; BMI)



9 (HDL-C; TG; SBP; ALT; HbA1c; LDL-C; HR; BMI; DBP)



We note that in the work by *Nair et al. (Nature medicine 2022)*, nine features were selected based on expert consensus regarding clinical relevance to T2D. In our feature selection strategy integrated both machine learning and deep learning approaches, yielding a data-driven and reproducible set of ten features. Interestingly, eight out of the ten features we identified overlapped with those reported by Nair et al., while the inclusion of two additional features (heart rate and LDL-C) may further capture population-specific patterns relevant to the Chinese cohort.

Thank you for your question, the decision to use ten features was not arbitrary but was based on systematic evaluation of classification performance, representation ability, tree structure robustness and also the clinical relevance to T2D.

We have clarified **Methods** section to address your concern.

3. Could you explain the X and Y axes of the DDRTree in the Methods section and the linear regression analyses of each phenotype and outcome against the tree dimensions?

Response: Thank you for your insightful question.

The DDRTree is a graph-based dimensionality reduction method that projects high-dimensional clinical data into a two-dimensional space, represented by the X and Y axes. These two axes do not have pre-defined biological meanings but instead serve as latent coordinates that preserve both local and global structures of the data manifold. Individuals with similar clinical features are positioned closer together on the tree, and the branching structure captures the continuum of phenotypic variation in type 2 diabetes.

To examine how specific variables shaped the tree structure, we performed linear regression analyses of each input variable (used to construct the tree) against the X and Y dimensions of the tree. This allowed us to quantify the influence of each variable on an individual's position within the embedded tree space. The resulting regression coefficients and 95% confidence intervals (CIs) are displayed in Figure 2D (new figure 4b). Variables with larger coefficients had a greater influence on tree topology.

Similarly, for outcome risks, we regressed the predicted probabilities (derived from cause-specific hazard models for each complication) against the tree dimensions to visualize their spatial distribution across the tree.

We have now clarified this explanation in the revised **Methods** section to help reader understanding.

4. As HbA1c is a key indicator for T2D diagnosis and monitoring, its low importance in the model could be discussed further. The authors may consider whether this reflects ethnic differences or potential biases.

Response: Thank you for noticing this question~ Interestingly, we observed that HbA1c, despite being a key biomarker for the diagnosis and monitoring of T2D, demonstrated relatively low importance in determining the tree topology in both our Chinese-specific model and the original Scottish tree model (*Nair et al., 2022*). This finding may initially

seem counterintuitive. However, several plausible explanations exist. First, because HbA1c is one of the defining criteria for T2D diagnosis, the range of values at the time of diagnosis tends to be relatively constrained, which reduces its discriminative power for phenotypic differentiation within a population already classified as having T2D. In contrast, other metabolic variables such as lipids or blood pressure often show greater variability and may therefore better capture the heterogeneity relevant to long-term complications.

Moreover, this observation could reflect population-specific pathophysiology. Prior studies have shown that East Asian populations, including Chinese individuals, often exhibit early β -cell dysfunction with relatively modest hyperglycemia and lower insulin resistance compared to Caucasians (*Ratnasingan et al., 2025; Ma & Chan, 2013*). This may result in less pronounced differences in HbA1c at diagnosis but greater variability in other metabolic features such as triglycerides and ALT, which emerged as more influential in our model.

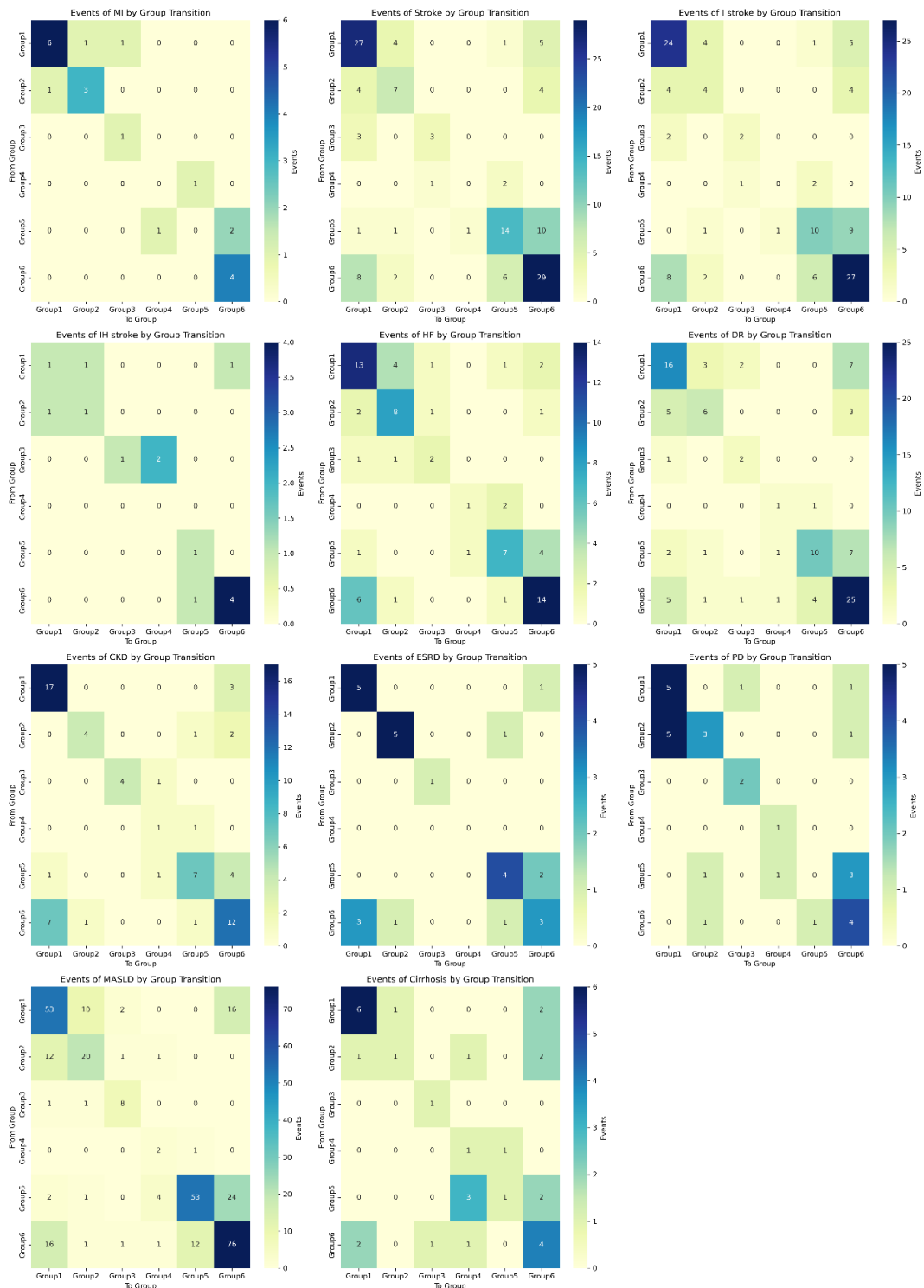
Together, our findings underscore the importance of moving beyond glycemia alone to capture the full spectrum of T2D heterogeneity, particularly when designing models for risk stratification and individualized care in non-European populations.

We have added in the revised **Discussion section**.

5. Individuals shifted their positions after five years. It may be helpful to conduct a supplementary analysis to examine whether those who moved into the high-risk group indeed experienced more complications.

Response: We thank the reviewer for this insightful suggestion. To address this, we conducted a supplementary analysis examining complication events among individuals who shifted between branches over the five-year follow-up period ($n = 1,472$). In this analysis, we primarily categorized patients into six phenotype-based groups (Group 1 to Group 6) based on the coordinates of the Chinese phenotype tree. Group 7 represents

a transitional or mixed cluster located near the center of the tree structure, which was not included in the main subtype definitions. Therefore, in interpreting the longitudinal branch-shift patterns and their associated complication risks, emphasis should be placed



on shifts among Groups 1 to 6.

As presented in **Supplementary figure 15**, we found that individuals who transitioned

into higher-risk phenotypic branches—particularly into branches 6—exhibited a noticeably higher number of diabetes-related complications, including myocardial infarction, stroke, diabetic retinopathy, chronic kidney disease, and liver-related outcomes. For example, among individuals who moved from Group 5 to Group 6, 10 experienced stroke and 24 developed MASLD, while those who stayed within Group 5 had fewer events. These findings provide further support for the clinical relevance of the phenotypic branches and their utility in tracking progression and guiding individualized care.

6. Could the authors explain how the influence of the variables on the tree coordinates was determined in the ‘development and performance of the Chinese tree model’ section (Figure 2D)?

Response: Thank you for your question same as question 3. As described in the Methods and shown in Figure 2D, we assessed the influence of each variable on the tree coordinates by performing linear regression analyses. Specifically, we regressed each individual phenotype (i.e., the ten variables used to construct the Chinese tree) against the two-dimensional tree coordinates (X and Y), derived from the reversed graph embedding. The regression coefficients, along with their 95% confidence intervals, were then plotted to indicate the direction and strength of association between each variable and the tree axes.

This approach allows us to quantify the extent to which each variable contributed to the spatial distribution of individuals within the tree structure. The results, displayed in Figure 2D, highlight the relative importance of the variables in shaping the topology of the tree.

7. The authors used Fine-Gray model and sub hazard ratios (sHRs) with death as a competing event for the main analysis instead of a sensitivity analysis. Please elaborate.

Response: We thank the reviewer for this insightful comment. The decision to use the Fine-Gray model with subdistribution hazard ratios (sHRs) as the main analytical approach was motivated by the nature of the clinical outcomes under investigation.

The primary outcomes in our study: diabetic retinopathy (DR), cardiovascular disease, kidney disease, and liver disease, are all subject to competing risk from death, particularly in older individuals or those with comorbid conditions. In the presence of competing risks, conventional Cox proportional hazards models may overestimate the incidence of the event of interest, as they treat competing events as censored.

The Fine-Gray model, on the other hand, allows for direct modeling of the cumulative incidence function and provides sHRs that more accurately reflect the real-world probability of experiencing the outcome, accounting for the presence of competing events.

Furthermore, 88% of our death data were derived from the Chinese CDC registry, which is a highly reliable source. This ensured the accuracy and completeness of death ascertainment in our cohort and further supported the use of a competing risk framework. Notably, prior studies such as the work by *Nair et al.* have also adopted Fine-Gray models as the primary approach when evaluating outcomes with substantial competing risks from mortality. Therefore, we considered this model more appropriate for our primary analysis. In contrast, while death information was available in the JADE cohort and thus allowed for the application of a competing risk model, such data were not available in the DR cohort. As a result, we applied a standard Cox proportional hazards model in the DR cohort, since the Fine-Gray approach requires reliable data on competing events. This choice was made to ensure internal consistency and model validity within each dataset.

We have clarified this justification in the **Methods section** (Pages 28-29, lines 690-718; Pages 30, lines 735-748) of the revised manuscript.

Minor comments:

8. ‘For this study, we included individuals aged 18 years or older from the CRDS who were diagnosed with type 2 diabetes (T2D; ICD-10 code E11) within the 12 months prior to baseline. Baseline was defined as the first recorded instance of a T2D diagnosis in the database. It is a bit confusing about the time for baseline and T2D diagnosis.

Response: We thank the reviewer for pointing out this lack of clarity. We agree that the original description could be misleading.

In this study, we aimed to include individuals with newly diagnosed T2D to better capture early disease features and minimize confounding from longer disease duration. To this end, we defined baseline as the first recorded T2D diagnosis (ICD-10 code E11) in the CRDS database.

To ensure that these were truly newly diagnosed cases, we retrospectively reviewed patients' medical records and prescription histories to confirm that the diagnosis had been made within the preceding 12 months. Therefore, the phrase “diagnosed within the 12 months prior to baseline” refers to individuals whose first T2D record in the database also reflected a confirmed clinical diagnosis within the prior year.

We have revised the **Methods section** (Page 21, lines 515-518) to clarify this definition.

9. Please use JADE cohort throughout the manuscript and the supplement instead of Hong Kong dataset in some of the tables and figures.

Response: We thank the reviewer for this helpful comment. We have carefully reviewed the manuscript and all supplementary materials, and have now replaced all instances of “Hong Kong dataset” with “JADE cohort” to ensure consistency throughout the text, tables, and figures.

10. Figure S15 was not mentioned in the manuscript.

Response: We appreciate the reviewer for pointing this out. We have now revised the manuscript to include a citation to supplementary materials at the appropriate location

in the **Results section** to ensure completeness and clarity. We made substantial revisions to both the main manuscript and the supplementary materials. If you find any part still unclear or needing further adjustment, we would be more than happy to make additional improvements. We sincerely appreciate your valuable advice.

Reviewer #1 (Remarks to the Author):

Overall summary

The authors have made substantial changes to the present manuscript. The authors, in addition, tried to address the feedback provided. However, in some cases the questions raised by the previous reviewers have not been directly addressed. In addition, the extensive revisions have also raised new questions and, in some cases, created additional issues that need clarification.

Therefore, my revised revisions are structured in the following way. First, general concerns, followed by a point-to-point evaluation of the responses to my previous comments (numbered from 1 to 15). Then, new comments based on the direct revisions provided from the first version of the paper to the current one, numbered from 16 to 45.

Response: We sincerely thank you for the detailed evaluation and constructive feedback on our revised manuscript. We greatly appreciate the time and effort taken to review our work and we acknowledge that.

We are also grateful for your continued attention to several issues raised previously, particularly regarding the criteria for population selection and related methodological considerations. These aspects were carefully determined from a clinical perspective, as both of our corresponding authors are practicing clinicians. We are truly pleased to have the opportunity to engage in this valuable exchange with you on such important clinical and methodological points.

Importantly, this review process has also been a valuable learning experience. Through your professional insights and our team's clinical background, we believe this exchange has deepened our mutual understanding across disciplines and enhanced the overall rigor and clarity of our work.

We deeply appreciate your thoughtful and meticulous review. We take your comments seriously, when there are errors, we correct them; when there are shortcomings, we strive to improve. This process has allowed us to refine our study.

In the following, we provide detailed responses to each of your comments. Where appropriate, we have revised the manuscript to ensure that the rationale, methods, and results are more clearly presented, and that any ambiguities are resolved. We hope these clarifications satisfactorily address your concerns.

General concerns

Despite substantial revisions, the manuscript remains difficult to evaluate due to unresolved methodological ambiguity and new inconsistencies introduced during revision. My overarching concerns are:

1) Coherence and consistency

The described pipeline, figures, and methods do not align (e.g., discrepancy between Fig. 1 and the narrative about how VAE outputs feed downstream steps; unclear group definitions in Fig. 6f; table totals that exceed the stated number of outlier individuals). Several minor errors (typos, repeated sentences, and a mislabeled loss term) further complicate interpretation (see comments 2, 8, 26, and 39–45).

Response: We sincerely thank you for highlighting the inconsistencies between the described workflow and figures. We have revised the manuscript for coherence and technical accuracy. Specifically, Figure 1 was redrawn to correctly depict that the VAE was used solely for feature attribution, and that downstream analyses were based on the original (non-reconstructed) variables. Figure 6f and the outlier (SupFig1) table have been corrected. Minor typographical and labeling errors, including a repeated sentence, the mis-specified loss term, and mislabeled encoder/decoder components, which have been corrected throughout the manuscript. All related figure legends and method descriptions have been aligned for consistency.

2) Role and justification of the VAE

It remains unclear whether the VAE is used for representation learning, feature selection, denoising, or all three, and why this choice is preferable to simpler alternatives (e.g. a vanilla autoencoder, PCA, or purely supervised selection). Assertions about

“informative” latent structure and “denoised input” are not supported by comparative analyses (see comments 3, 10, 14, 18, 27–28, 30–33, and 36).

Response: We appreciate the opportunity to clarify the rationale for incorporating the VAE. In this study, the VAE serves as a nonlinear feature extraction tool to identify key variables representing metabolic heterogeneity in T2D. The focus of our work is on the ten variables prioritized by the VAE, which we subsequently evaluated using independent supervised-learning analyses. We have clarified this conceptual framework in the revised Methods and Discussion sections. Corresponding claims about “denoising” or “informative latent structures” have been applied to better reflect the evidence.

3) Evaluation design, hyperparameters, and reproducibility

The manuscript lacks essential details on hyperparameter search spaces, selection criteria, and safeguards against data leakage. Model performance is reported without clear nested CV, fold definitions. It is also unclear when models use the full feature set versus the 10 “VAE-selected” features (see comments 10–12, 25–26, 33–35, 38). The proposed “stability” metric based on cosine similarity across independently trained VAEs is not valid given rotational/permutation of latent spaces (see comment 24).

Response: We agree that methodological transparency and reproducibility are critical. Accordingly, we have substantially expanded the Methods section to specify all hyperparameter settings, tuning procedures, and validation design. Each supervised model was evaluated using 10-fold cross-validation implemented via `cross_val_score` in scikit-learn, with consistent random-seed control. LASSO and ElasticNet used `LogisticRegressionCV` for automatic regularization tuning and other classifiers were run with standard parameters to ensure comparability. We now explicitly state in the revised manuscript which models used the full feature set and which used the ten VAE-selected features, and provide a direct comparison of both in Supplementary Fig. 5. The cosine-similarity “stability” metric has been revised. Finally, we have made all scripts

and preprocessing workflows publicly available on GitHub for full reproducibility (Please see <https://github.com/Tongyue1999/Chinese-DDrtree>).

4) Use of controls and inclusion/exclusion criteria

The inclusion of non-diabetic controls in representation learning is not sufficiently justified if the stated goal is to extract diabetes-specific patterns. The chosen case:control ratio is also not motivated. On the other hand, exclusion of multimorbid individuals may limit real-world applicability and should be addressed (see comments 5, 9 and 21).

Response: We thank you for this thoughtful comment. We have clarified that non-diabetic controls were included only during the VAE representation-learning stage to stabilize the feature space and capture the T2D physiological spectrum, but all subsequent analyses—including DDRTree construction and subgroup identification—were performed exclusively within the T2D cohort. The case-to-control ratio was determined by the data availability of the real-world dataset rather than by design. Regarding exclusion criteria, we have elaborated that individuals with severe multimorbidity were excluded to minimize confounding and to focus on early-stage T2D heterogeneity. From a clinical and practical perspective, we believe that this exclusion criteria more accurately reflect our research objective, which is to delineate the intrinsic metabolic heterogeneity of newly diagnosed T2D. We also now explicitly acknowledge that external validation in multimorbid populations is warranted in future studies.

5) Claims that exceed current evidence

Statements about longitudinal “phenotypic shift,” superiority of DDRTree over traditional clustering, mitigation of confounding via VAE-based selection, clinical relevance of the continuous model, and “early interventions” are insufficiently supported by statistical testing, benchmarking, or appropriate references (see comments 16–17, 20, 28–29, and 41–42).

Response: We appreciate your caution regarding interpretation. We have moderated the language throughout the manuscript to ensure that conclusions are commensurate with the supporting data. References to “significant longitudinal phenotypic shift,” “superior performance of DDRTree,” “confounding mitigation,” and “early intervention potential” have been rewritten to describe observed trends and methodological advantages rather than confirmed clinical outcomes. We removed the term “significant” when formal tests were not performed. The revised text emphasizes that the present study provides an exploratory framework for understanding T2D heterogeneity and does not claim demonstrated clinical efficacy or intervention effects.

Point-by-point evaluation of responses

Comment 1: In my previous comment #1 I addressed inconsistencies in the model pipeline in the abstract. The authors have revised this part adequately.

Response: Thank you for your reply.

Comment 2: The authors addressed my previous comment #2 by providing more context in the introduction. However, their introduced Figure 1 does not agree with the methodology described elsewhere in the manuscript. In particular, in “STEP2: Selecting T2D Features”, it appears if the reconstructed input from the variational autoencoder (VAE) is fed into the machine learning selection pipeline. Instead it should (probably?) reflect that the VAE is only used for feature selection based on feature attribution to the latent representations.

Response: We sincerely thank you for this insightful comment. We agree that our original Figure 1 may have unintentionally implied that the reconstructed input from the VAE was used for downstream machine learning. In fact, as you have correctly pointed out, the VAE was only used to derive feature attributions based on the latent representations. These attributions were then used to guide the subsequent machine learning–based feature selection, without feeding reconstructed inputs into the pipeline. We have revised Figure 1 accordingly to more clearly illustrate that: the VAE was

applied to evaluate the contribution of each original clinical feature to the latent data structure. Based on this unsupervised feature assessment, ten variables were selected as the most informative. These top-ranked original features were subsequently included in the machine learning step to confirm the robustness of the selected features.

The revised figure now accurately reflects the described methodology.

Comment 3: In my previous comment #3 I noted the authors that the VAE is a generative model, not necessarily guaranteed to provide informative representations for predictive modelling. The authors replied that in fact it is solely is used for feature extraction, and that its “design allows for a continuous and smoother latent space, making VAE not only more generative in nature, but also better suited to capture the intrinsic structure of the input data”. When making these statements regarding the latent space, it appears as if the authors compare the VAE approach to other dimensionality reduction techniques (such as e.g. a vanilla AE, a denoising AE, PCA, or other generative models), but this is not the case. Following, the authors write in their response “While reconstruction is the primary training objective, we interpret this bottleneck as a form of feature-preserving compression, where essential input characteristics are retained to ensure accurate reconstruction. As such, the reconstructed input (X') remains close to the original input (X), validating that the learned representations preserves relevant information”. There are two things to this comment. Firstly, a good reconstruction does not validate that the representations preserves ‘relevant’ information. The VAE can simply overfit to a ‘dictionary’ remembering the input data from the latent representations, without the latent representations having any meaning (see e.g. Alain, Guillaume, and Yoshua Bengio. "What regularized auto-encoders learn from the data-generating distribution." *The Journal of Machine Learning Research* 15.1 (2014): 3563-3593.). Secondly, given the authors focus on the reconstruction error, why not just use the (much simpler) vanilla autoencoder? Why add the complexity of the VAE? Later in the authors response they touch upon the nonlinear capabilities of the VAE, which are desirable for modelling complex correlations, but in

no way does it guarantee “denoised input” or “informative patterns”.

Response: We sincerely appreciate your insightful comments and fully agree that a low reconstruction error alone does not imply that the latent representations capture biologically meaningful or predictive information.

And we thank you for highlighting the important discussion by Alain and Bengio (2014) regarding the interpretability of autoencoder representations. We fully acknowledge that, in the absence of appropriate regularization, an autoencoder (including its variational form) may overfit by effectively memorizing the training data rather than learning meaningful latent structures. We also note that the page range of the cited article was incorrectly stated in your comment; the correct reference is: Guillaume Alain and Yoshua Bengio. Journal of Machine Learning Research 15(110): 3743–3773, 2014.

This principle has been successfully applied in prior work using VAE-based frameworks for multi-omics data integration and feature discovery (*Nat Biotechnol* 41(3):399-408, 2023).

In our setting, the VAE latent space was used to derive feature attributions, identifying the most informative clinical variables contributing to the shared latent structure-rather than relying on reconstruction loss or data generation. This approach enables extraction of robust, low-dimensional representations that preserve the underlying relational patterns within the data, thereby supporting downstream predictive modeling and interpretation.

We have accordingly revised the Methods sections to clarify this conceptual focus, removing the misleading wording about “denoising” or “informative patterns”.

Comment 4: The authors clarified and addressed my previous comments adequately.

Response: Thank you for your acknowledgement.

Comment 5: My previous comment #5 addressed a lack of justification for individuals with major medical conditions being excluded. The authors responded saying the

excluded to “reduce potential confounding effects that could distort the identification of T2D subgroups”. It is true that individuals with many complicated comorbidities would likely alter the biomarker profiles used in the clustering framework. On the other hand, many individuals living with diabetes are multimorbid (often upwards of 33%, see ref Khunti, K., Chudasama, Y. V., Gregg, E. W., Kamkuemah, M., Misra, S., Suls, J., ... & Valabhji, J. (2023). Diabetes and multiple long-term conditions: a review of our current global health challenge. *Diabetes Care*, 46(12), 2092-2101), so it is true that inclusions of comorbidities could bias diabetes-specific phenotypic patterns, but it also excludes potential usefulness of the derived model in the real world where multimorbid diabetes patients are prevalent.

Response: We thank you for this thoughtful and important comment. We fully agree that multimorbidity is common among individuals living with established diabetes, as emphasized in the cited references (*Diabetes Care*.46(12):2092-2101, 2023. DOI: 10.2337/dci23-0035; *BMC Med.* 25(17):145 ,2019. DOI: 10.1186/s12916-019-1373-y). Our rationale for excluding individuals with major medical conditions at baseline was to minimize potential confounding effects that could obscure the intrinsic heterogeneity of T2D at diagnosis, in other words, at early stage of T2D. Specifically, comorbidities such as chronic kidney disease, advanced liver disease, or malignancy can substantially alter metabolic and biochemical profiles, thereby masking diabetes-specific pathophysiological patterns.

This design choice aligns with prior studies exploring diabetes heterogeneity (e.g., Nair et al., *Nat Med* 28: 982 – 988, 2022; *Lancet Diabetes Endocrinol* 12(2):119-131, 2024; and *Diabetes* db250197, 2025 DOI: 10.2337/db25-0197), which similarly focused on **newly diagnosed T2D populations** to ensure that clustering reflects endogenous disease mechanisms rather than secondary metabolic effects of comorbidities or treatments. From a clinical and practical perspective, we believe that this exclusion criterion more accurately addresses our research question, which aims to capture the intrinsic metabolic heterogeneity of newly diagnosed T2D.

We also acknowledge your important point that multimorbid diabetes is common in

real-world clinical practice. While our current analysis aimed to delineate the baseline pathophysiological structure of T2D without major external influences, future work will extend this framework to broader, multimorbid populations.

Comment 6: The authors clarified and addressed my previous comments adequately.

Response: Thank you for your kind comment.

Comment 7: The authors clarified and addressed my previous comments adequately.

Response: Thank you for your kind comment.

Comment 8: My previous comments address the complex data preprocessing steps the authors apply in their modeling pipeline. The authors responded with clarification on the steps. In particular, the authors provide a table with individuals with outlier data (in total 575 individuals), but the numbers in the table are larger than 575 so I am unsure what to make of the table. Could the authors please clarify?

Response: We thank you for pointing this out. In the earlier version, the total number of individuals with outlier data was mistakenly written as 575 due to a typographical error. The correct number is 1904 individuals with at least one variable flagged as an outlier. Since one individual can be counted in more than one variable, these numbers are not additive. After accounting for overlaps, a total of 1904 individuals had ≥ 1 variable identified as an outlier. The distribution was as follows: 1850 had one variable affected, 52 had two, 2 had three, and 0 had four or more.

Number of variables with outliers	Number of individuals
1 variable	1850
2 variables	52
3 variables	2
≥ 4 variables	0

Total	1904
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Also, we have corrected the flowchart in supplementary figure1.

Comment 9: My previous comment 9 raised concerns regarding why controls were sampled for the representation learning part of the model pipeline. The authors replied: “The inclusion of controls was intended to enhance the contrast between diabetic and non-diabetic phenotypes, allowing for better identification of diabetes-specific patterns, particularly during unsupervised learning and latent space evaluation”. I do not believe that answers my question. Can the authors elaborate? Adding the non-diabetic individuals is rather counterintuitive IF the goal is diabetes-specific patterns. So-called “enhanced contrasts” will distort the diabetes-specific patterns, not enhance them. If applying the authors logic then I see no reason to exclude multimorbid diabetes patients, as those could then also “enhance contrasts”.

Response: We sincerely thank you for raising this important point. In our analysis, non-diabetic controls were not included in the DDRTree clustering process. Their inclusion was limited to the representation learning stage of the VAE model, where they served to regularize and stabilize the learned latent space by providing reference anchors for the full physiological range of metabolic variation. This design choice aimed to improve the robustness of latent representations by preventing overfitting to diabetes-only distributions.

Specifically, the VAE was trained to identify the most informative clinical variables associated with T2D subtypes in the Chinese population, to evaluate the contribution of each original clinical feature to the latent data structure. Based on this unsupervised feature assessment, ten variables were selected as the most informative. We used the original (non-reconstructed) values of these ten variables as input subsequently used to construct the Chinese-specific diabetes tree.

As you correctly note that, the purpose was not to enhance “contrast” between disease states, but rather to normalize the feature space in which diabetes-related deviations could be more reliably identified. We have revised the corresponding paragraph in the

Figure1, Methods sections to clarify this rationale.

Comment 10: In my previous comment #10, I urged the authors to justify the choice of VAE, as opposed to simpler alternatives. The authors have attempted to adhere to my comments by adding a LASSO and elastic net model (showing the performance of the model), yet it is unclear how the authors implemented these models. Did they let apply them on the full dataset or the subset of 10 features as the other ML model (i.e. the GBC). In general, it would be useful to compare the performance of the ML models when using the 10 “VAE-selected” features versus using the full featureset. Also, the manuscripts remains with no details regarding which hyperparameters were chosen in the supervised machine learning models. For example, in the LASSO model the lambda parameter controls the L1 shrinkage penalty. Ideally, these should be tuned, and the total performance evaluated in a nested CV setup.

Response: We thank you for this valuable and detailed comment.

One of our objectives was to identify the most informative clinical variables associated with T2D in the Chinese population. To this end, we employed a VAE to quantify the contribution of each clinical feature to the latent data structure, and selected the ten most informative variables.

To validate their discriminative reliability, we first applied multiple supervised classifiers—including Logistic Regression, Random Forest, Gradient Boosting, KNN, SVM, Decision Tree, LASSO, and Elastic Net—using the full feature set (Supplementary Fig. 5a, b). Among these, the Gradient Boosting Classifier (GBC) achieved the highest performance, followed by Random Forest (RF).

We then focused on these two best-performing models and re-evaluated them using both the full feature set and the VAE-selected subsets containing the top 5 and 10 features (Supplementary Fig. 5c). This analysis demonstrated that the reduced feature sets (10 variables) maintained comparable predictive performance to the full set, confirming the robustness and representativeness of the VAE-selected variables.

All models were implemented in Python using scikit-learn. LASSO and Elastic Net

used LogisticRegressionCV (10-fold) to automatically tune the regularization strength (λ , the inverse of C). Other classifiers were trained with standard scikit-learn hyperparameters (e.g., `n_estimators = 100`, `learning_rate = 0.1` for GBC) for fair comparison. Model performance was evaluated using 10-fold cross-validation (`cross_val_score`), and the average accuracy across folds was reported.

Although we did not perform a fully nested cross-validation—as our goal was validation rather than predictive optimization—the consistent ranking of models across folds supports the robustness of the comparative evaluation. Importantly, the machine-learning analysis was designed solely to validate the reliability and discriminative power of the VAE-derived features, not to optimize predictive performance. We have clarified this in the revised Methods and Supplementary Figure legends.

Comment 11: My previous comment #11 addressed the lack of detail in describing the methods. The authors have done some work to improve the manuscript in this regard, but there are still details and justifications lacking (see additional comments)

Response: We thank you for raising this point. We have further revised the Methods section to provide additional implementation details and justifications. All supervised machine learning models were implemented in Python using scikit-learn. Except for LASSO and Elastic Net, which employed LogisticRegressionCV for automated regularization tuning, all other classifiers were trained using the default hyperparameters of scikit-learn (e.g., `n_estimators=100` for Random Forest, `learning_rate=0.1` and `max_depth=3` for Gradient Boosting, `kernel='rbf'` for SVM) to ensure reproducibility and comparability across models. Model performance was evaluated via 10-fold cross-validation, with results quantified using multiple metrics, including accuracy, ROC AUC, and PR AUC, to ensure robustness across potentially imbalanced data. Among all classifiers, GBC achieved the highest performance (ROC AUC > 0.9), followed closely by RF, and was therefore selected as the optimal model for downstream validation. To further assess the stability of feature selection, we reconstructed classification models using the top 5–10 VAE-derived features and compared their predictive performance with those using the full feature set

(Supplementary Fig. 5). The comparable performance confirmed that the 10 VAE-selected features were sufficient and reliable for characterizing T2D heterogeneity. Importantly, as clarified in the revised manuscript, the primary aim of this analysis was not to optimize predictive modeling but to validate the robustness of VAE-derived features in representing T2D heterogeneity within the Chinese population. We have revised the methods according to your suggestion.

Comment 12: My previous comment #12 concerned the reporting of ML performance and the metrics used. The authors have partly addressed these comments by including the ROC AUC and PR AUC for their assessment, yet I wonder how hyperparameters were chosen when making these assessments (see comment 10).

Response: We thank the reviewer for this point.

In our analysis, we aimed to compare the overall discriminative capacity of different classifiers rather than to optimize each model's predictive performance. Therefore, all models were implemented with standardized configurations to ensure fair comparison and reproducibility (e.g., `n_estimators=100` for Random Forest, `learning_rate=0.1` for Gradient Boosting, `kernel='rbf'` for SVM). LASSO and Elastic Net regularization were implemented using `LogisticRegressionCV` (10-fold) in `scikit-learn`, which automatically tunes λ (the inverse of C). All classifiers were trained and evaluated using 10-fold cross-validation (`shuffle=True`, `random_state=42`), and model performance was summarized using accuracy, ROC AUC, and PR AUC.

This consistent configuration allows fair evaluation of the discriminative value of the clinical variables. We have clarified this rationale and provided the corresponding code at the project GitHub (<https://github.com/Tongyue1999/Chinese-DDrtree>).

Comment 13: The authors clarified and addressed my previous comments adequately.

Response: Thank you for your reply.

Comment 14: My previous comment #14 concerned the feature importance measure

based off the VAE weights. The authors have partly addressed these comments. Yet, the method is still poorly presented mathematically in the methods section. The authors write in their reply: “A weight-based approach, computing the sum of absolute encoder weights from each input to all hidden nodes”. I wonder if this applies to both the means and the variances in the hidden layer, or solely the means?

Response: Thank you for your reply. We appreciate your careful reading and for pointing out the need for a more precise mathematical description of our feature importance procedure.

Our weight-based importance is defined on the encoder trunk (input→hidden) only and does not include the parameters of the mean (μ) or variance ($\log\sigma^2$) heads. With one hidden layer (512 units) and latent size 32, letting $W^{(1)} \in \mathbb{R}^{512 \times D}$, we report

$$I_f^{\text{weight}} = \sum_{h=1}^{512} |W_{h,f}^{(1)}|.$$

This metric uses only shared encoder weights, avoiding scale and noise-sensitivity differences between the mean and variance heads, thus providing a stable measure of input feature influence. Complementarily, we report a perturbation-based metric (testing encoder mean changes under feature masking) to directly assess functional impacts on latent representations. We have revised the Methods section to explicitly specify that the weight-based importance.

Comment 15: The authors clarified and addressed my previous comments adequately.

Response: Thank you for your reply.

New comments based on revisions made.

Major comments

Comment 16: In the abstract, a “significant longitudinal phenotypic shift” is described. Yet, in the manuscript no formal statistical tests are performed. Even if performed it is difficult to translate those into any meaning, as they are based on latent dimensions

from a dimensionality reduction method.

Response: We thank you for this valuable comment. We agree that the term “significant” might be misleading, as the observed changes are derived from latent dimensions of the model rather than direct statistical comparisons. Therefore, we have removed “significant” from the abstract and revised the sentence as follows: This Chinese tree model was validated in two independent external cohorts and revealed longitudinal phenotypic shift trending towards higher-risk branches.

Comment 17: In the introduction, the authors have added the sentence “This flexible framework has also been successfully applied to type 1 diabetes and prediabetes, where it enabled the delineation of heterogeneous disease trajectories and facilitated the identification of early interventions tailored to specific risk patterns”, followed by two references. The two referenced studies do not directly facilitate nor evaluate so-called “early interventions”. They simply briefly discuss that such a model could be used for design of such interventions. There is a fundamental difference between using risk prediction models for providing information for e.g. a GP and risk prediction models designed specifically with an intervention in mind (see e.g. van Geloven, Nan, et al. "The risks of risk assessment: causal blind spots when using prediction models for treatment decisions." *Annals of internal medicine* (2025)).

Response: We thank you for this important point. We agree that the cited studies did not directly evaluate or implement early interventions, but rather discussed their potential implications for intervention strategies. To avoid overstatement, we have revised the sentence as follows: This flexible framework has also been applied to type 1 diabetes and prediabetes, where it enables the delineation of heterogeneous disease trajectories and informs the design of early interventions tailored to specific risk patterns^{13, 14}.

Comment 18: In the introduction, the authors have added the sentence: “While the learned features are not specifically optimized for downstream predictive modeling tasks, they can still reveal meaningful patterns and help identify clinically relevant

variables when conventional selection methods are insufficient.” While this is true, the authors have not argued for why these conventional selection methods are insufficient in their setting, nor have they looked at e.g. differences in selected variables if one employs a “traditional” selection method, compared to their complex VAE-based setup.

Response: We appreciate your thoughtful comment. We agree that our current analysis does not include a direct comparison between the VAE-based and the traditional feature selection methods. Our intention was to emphasize the potential of the VAE framework to capture nonlinear and higher-order dependencies among variables, which may not be well represented by conventional linear or univariate selection approaches. To clarify this point, we have revised the sentence as follows: While the learned features are not specifically optimized for downstream predictive modeling tasks, they can still reveal meaningful patterns and highlight potentially relevant clinical variables, particularly in complex datasets.

Comment 19: In the “Mapping the Chinese individuals onto the Scottish tree model” section, the authors write “a distance-based algorithm assigned each individual to the nearest node on the Scottish reference tree.” The authors should be specific in which algorithm (or metric) is used.

Response: We thank you for this helpful suggestion. We have clarified this in the revised manuscript. Specifically, we used a distance-estimating algorithm that calculates the Euclidean distance between each Chinese individual and all nodes in the two-dimensional latent space of the Scottish reference tree, assigning each individual to the nearest node based on the minimum Euclidean distance.

Comment 20: In the same section, the authors have now clarified “For each outcome, we applied Fine–Gray competing risk models, using death as a competing event. Ten-year event probabilities were estimated for each individual based on their coordinates in the two-dimensional tree space.” Were these predictions based solely on the

coordinates? Why not control for sex and age, to ensure that differences in estimates are not driven entirely by differences in baseline characteristics by the patients.

Response: We thank you for this important comment. As detailed in the Methods, we developed three versions of the Fine-Gray models: Using only the DDRTree-derived tree dimensions; Incorporating the DDRTree dimensions with spline functions; Fully adjusted models including age, sex, ten selected clinical characteristics, and duration of T2D. These progressively adjusted models allow us to account for baseline differences, ensuring that the observed outcome patterns are not driven solely by age, sex, or other covariates.

Comment 21: In the “Development and performance of the Chinese tree model” section, the authors write “The VAE model was trained using data from 45,589 individuals with type 2 diabetes (T2D) and 11,340 healthy individuals (without diabetes).” This choice (both in terms of why healthy individuals are included, but also the ratio of T2D to controls) should be justified, and it is not currently so in the present manuscript.

Response: We thank you for this important comment. The inclusion of healthy individuals in the VAE training was intended to provide a broader context for learning latent representations, allowing the model to better distinguish disease-specific patterns from the background of normal variation. This can help the latent space capture biologically meaningful features that differentiate T2D from healthy states.

Regarding the ratio, the available healthy control cohort was limited to 11,340 individuals, resulting in a T2D-to-control ratio of approximately 4:1. While this ratio could not be altered due to data availability, all features were standardized during training, and the downstream analyses (feature selection and Chinese tree construction) were based solely on the T2D individuals, ensuring that model outputs and interpretations reflect diabetes-specific patterns.

Comment 22: In the same section, the authors mention “we evaluated both the reconstruction accuracy of the input data and the model’s stability across multiple

refitting procedures”. It is not clear what is meant by “the model’s stability here”.

Response: We thank you for this comment. We have clarified in the revised manuscript that model stability was assessed by refitting the VAE multiple times using the same hyperparameters and comparing the resulting latent representations.

Comment 23: In the same section, the authors mention a reconstruction accuracy ranging from 0.85 to 1.0 which seems very high. A perfect reconstruction for a VAE could imply that a dictionary mapping was learned, and that the KL penalty factor in the ELBO were not enforced. How do the KL divergence loss term look in these cases? Have you visualized the latent representations for these “perfect” situations to ensure you have not just learned a dictionary.

Response: We appreciate your concern and have replaced the earlier coarse range with distributional summaries computed from our runs. Training-set reconstruction is quantified as cosine similarity between inputs and reconstructions for features, with the ELBO evaluated at full $\beta=0.001$. In a 54-configuration, multi-seed grid the reconstruction metric spans 0.532 – 0.846 (IQR 0.759 – 0.818; median 0.793), far from

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Comment 24: The authors use average change in cosine similarity “within the latent space” as a measure of stability. However, VAEs don’t have an inherent “coordinate system” in the latent space. If you train two VAEs on the same data (or different folds), their latent axes can rotate, flip, or permute without changing the learned distribution. So the same data point may map to quite different coordinates in the two spaces. That means cosine similarity between the raw latent vector of VAE A and VAE B is not guaranteed to reflect meaningful closeness.

Response: We fully agree with your observation that latent coordinates in VAEs are not inherently identifiable and can undergo arbitrary orthogonal transformations (e.g., rotation, flipping, permutation). To address this, we did not compare raw latent

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coordinates between different training runs.

Instead, we quantified model stability in the reconstruction space, which is invariant to latent coordinate rotations. Specifically, for each hyperparameter configuration, we trained the VAE five times independently on the same dataset. Using the first run as the *reference*, we computed, for each individual, the cosine similarity between its reconstructed feature vector from the reference run ($\hat{x}_{ref}$) and that from each of the other runs ($\hat{x}_r$). The average of $1 - \cos(\hat{x}_{ref}, \hat{x}_r)$ across all individuals was then used as the measure of reconstruction difference between runs-smaller values indicate higher stability.

This metric naturally avoids dependence on the latent coordinate system and directly evaluates the consistency of the encoder-decoder mapping across repeated trainings. We have clarified this description in the revised Methods section and figure legend. Importantly, this refinement does not alter the conclusions of the study.

Comment 25: In the same section, the authors write “We first trained several models using all input features and identified the Gradient Boosting Classifier (GBC) as the best-performing model based on 10-fold cross-validation.” How were hyperparameters for the GBC (and other algorithms) chosen in this evaluation setup? It is not clear if an internal optimization of hyperparameters was performed.

Response: We thank you for this comment. In our evaluation, all models, including the GBC, were trained using conventional default parameters in scikit-learn (as shown in the provided code on GitHub). To ensure fair comparison and reproducibility, all models were implemented using consistent configurations and standard hyperparameters (e.g., `n_estimators=100`, `learning_rate=0.1`, `max_depth=3` for Gradient Boosting). We acknowledge that further hyperparameter optimization could improve absolute performance, but this does not affect the main conclusions regarding the comparative performance of the classifiers on our dataset as we mentioned before.

Comment 26: The authors write “the accuracy had already exceeded 0.915, closely

approaching the performance of the full-feature model (Supplementary Fig. 5).” The full-feature model is not in supp fig 5, so it would be useful to contrast the performance of the ML algorithm based on the subset and based on the full features. This goes in general for all of the evaluated models.

Response: We thank you for pointing out this issue. We have revised the Results section to explicitly compare the performance of models trained on all features versus those using only the top-ranked variables identified by the VAE. Specifically, as now stated in the revised text: The Gradient Boosting model achieved an AUC of 0.926 using all features, which remained comparable (AUC = 0.916) when restricted to the top 5 to 10 variables, with similarly high PR-AUC and F1 scores (Supplementary Fig. 5).

These findings indicate that the top 10 features capture the majority of predictive information, with limited incremental value from additional variables. Supplementary Fig. 5 has also been updated to include both sets of results for clarity.

Comment 27: In the discussion the authors write “We then applied gradient-boosting classifiers to identify the top variables contributing to subtype discrimination.” This is confusing as it is not what is described in the rest of the manuscript. My understanding is that you selected features based on reconstruction importance in the VAE and/or shap values linking to their representations.

Response: We thank you for pointing this out. We have corrected the Discussion to clarify that feature importance was assessed using VAE encoder weights and SHAP-based perturbation analysis on the latent representations, not gradient-boosting classifiers. Feature importance was primarily derived from VAE encoder weights and SHAP-based perturbation analysis, ensuring that the selected variables reflect meaningful latent representations. The revised text now clearly reflects the methodology described in the Discussion section. We then trained a gradient-boosting classifier using the 10 VAE-selected features to verify that this subset retained predictive performance comparable to the full features.

Comment 28: The authors claim that “Importantly, this method helps mitigate the confounding effects of variable co-dependence, which may bias subtype classification in traditional models relying on manually selected features.” How exactly is this the case? A data-driven selection (via VAE) is less subjective than picking only “clinically relevant” variables – I agree with that. However, VAEs don’t remove variable codependence; they model it. The latent factors will still be shaped by correlated inputs. The choice of “most influential variables” is model-dependent (different seeds, hyperparameters, or architectures could change the ranking). Influence measures can still be biased toward variables with strong variance or measurement stability, not necessarily those with the deepest clinical meaning. Saying that the selection process mitigates confounding effects is too strong. It mitigates selection bias, but not confounding directly.

Response: We thank you for this important clarification. We have revised the manuscript to clarify that the VAE-based feature selection primarily reduces selection bias from manually chosen variables. The VAE models the co-dependence among variables but does not directly eliminate confounding; instead, it provides a reproducible framework to identify variables most informative for defining T2D subtypes, acknowledging that importance measures can be influenced by model initialization, hyperparameters, or input variability. We have revised the sentence according to your suggestion.

Comment 29: The authors write: “This large sample size enabled us to construct a robust two- dimensional tree structure using the DDRTree algorithm, which outperforms traditional clustering methods in capturing the complexity and continuity of T2D progression.” This is a claim. The authors have not benchmarked against traditional clustering methods nor referenced to studies who do.

Response: We thank you for pointing this out. We have removed the unsupported comparison and now describe the DDRTree structure without making claims relative to

other clustering methods.

Comment 30: In the “Design of VAE section” the authors write that the mean and log variances projected by the encoder defines a multivariate Gaussian distribution, but then they write $N(0,1)$, suggesting it be a multivariate standard normal. If so, how is the information from the encoder used? Based on the current description it is not clear.

Response: We appreciate the reviewer’s comment and agree that our wording could conflate the learned posterior with the standard normal prior. In our VAE, the encoder outputs two vectors, $\mu_\phi(x)$ and $\log \sigma_\phi^2(x)$, which parameterize a data-dependent approximate posterior over the latent variables:

$$q_\phi(z | x) = \mathcal{N}(\mu_\phi(x), \text{diag}(\sigma_\phi^2(x))).$$

Latent samples are obtained via the reparameterization trick,

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \varepsilon, \varepsilon \sim \mathcal{N}(0, I),$$

so the encoder’s outputs (μ, σ) directly determine the distribution from which z is drawn. The notation $\mathcal{N}(0, I)$ appears in two distinct roles: (i) as the distribution of the auxiliary noise ε used in reparameterization, and (ii) as the fixed prior $p(z) = \mathcal{N}(0, I)$ used solely in the KL regularization term. We do not assume the posterior is standard normal. We will revise the Methods to make this distinction explicit.

Comment 31: In the same section – the authors write that they use both ReLU and LeakyReLU as activation functions – could the authors clarify which one, and if both describe in which settings one is used instead of the other?

Response: We thank you for this comment. We clarify that only LeakyReLU activation functions are used in the encoder, not the ReLU. Thank you for your careful examination.

Comment 32: In the same section it is described that a flat 10% dropout and 32-dimensional latent space was used, yet you write in the manuscript (and previous

rebuttal letter) that you optimized over these.

Response: We thank you for this helpful comment and for catching this source of ambiguity.

In our study, we systematically optimized the VAE hyperparameters to identify the configuration that best captured the underlying data structure while maintaining generalization. Specifically, we tested multiple architectures with varying hidden layer sizes (256, 512, 1,024), number of layers (1–3), latent dimensions (24, 32, 48), dropout rates (10%, 20%, 30%), and KLD weights (1×10^{-3} , 1×10^{-4} , 1×10^{-5}). The final architecture comprising a single hidden layer with 512 neurons, a dropout rate of 10%, and a 32-dimensional latent space, which was selected based on its superior reconstruction accuracy and stability across validation runs.

We agree that the previous description might have implied that these settings were a priori fixed rather than the result of optimization. To avoid confusion, we have revised the Methods section to clarify that these parameters were the optimal configuration obtained after hyperparameter tuning, and have removed the redundant earlier phrasing that suggested otherwise.

Comment 33: In the “model training” section, the authors mention the evidence lower bound (ELBO), yet it is not described elsewhere in the manuscript. It’s relation to the loss function in the manuscript should be specified.

Response: We thank you for this comment. We optimize a weighted negative-ELBO. In our implementation, the training loss reported in the manuscript corresponds to the negative ELBO under a unit-variance Gaussian decoder, expressed as the sum of a reconstruction term and a KL regularization term:

$$\mathcal{L}_{\text{total}}(\theta, \phi) = \underbrace{\frac{1}{B} \sum_{n=1}^B \|x_n - \hat{x}_n\|_2^2}_{\text{MSE reconstruction}} + w \underbrace{\frac{1}{B} \sum_{n=1}^B \frac{1}{2} \sum_{j=1}^L (\mu_{n,j}^2 + \sigma_{n,j}^2 - 1 - \log \sigma_{n,j}^2)}_{\text{KL}(q_\phi(z|x) \parallel p(z))}.$$

With a Gaussian likelihood for the decoder, the MSE term equals the negative log-likelihood; hence minimizing $\mathcal{L}_{\text{total}}$ is equivalent to maximizing the ELBO.

Comment 34: In the hyperparameter optimization section, it is stated that the dataset was split 90/10. Since the authors are using this to guide model selection, I suppose this split was performed on the train dataset?

Response: We thank you for this comment. We clarify that we initially divided the dataset into a training set (90%) and a test set (10%) to identify the optimal hyperparameter settings to capture the data structure without losing the ability to generalize on the test data. We have revised the Methods section to clarify this point.

Comment 35: The authors write that performance was evaluated using both test log-likelihood and reconstruction accuracy. How was test log-likelihood assessed? It is not clear in VAEs how to do this.

Response: We thank you for this comment. Our original description of “test log-likelihood” followed the terminology used in *Nature Biotechnology* (2023, Vol. 41, pp. 399 – 410), whose methodology our work was initially based on. We acknowledge that this terminology was not fully accurate in the VAE context, as the log-likelihood is typically approximated via the Evidence Lower Bound (ELBO).

In our results, “test log-likelihood” refers to the batch-averaged VAE objective (negative ELBO) evaluated on the evaluation dataset, computed as the same loss used for training but with the full KL weight w . Concretely, we compute: Decoder likelihood assumption: unit-variance Gaussian, so MSE equals negative log-likelihood up to a constant; the total corresponds to a (weighted) negative ELBO. Lower is better and directly comparable across configurations under identical preprocessing and w . Evaluation dataset: by default, we evaluate on the training set (for relative model comparison); a held-out set can be evaluated by feeding it through the same scaler and loss computation. In our metrics runs, the evaluation used the training data to provide consistent, like-for-like comparisons between configurations.

Accordingly, we have revised the text for clarity.

Comment 36: The formula for the feature importance implies a sum over weights in just a single hidden layer, yet the authors also evaluate architectures with more than 1 hidden layer before the latent bottleneck. Can you generalize the formula to these cases as well.

Response: We thank you for this comment. For our single-layer encoder, feature importance is calculated as:

$$I_f^{\text{weight}} = \sum_{h=1}^H |W_{h,f}|$$

where $W \in \mathbb{R}^{H \times D}$ connects input features to the hidden layer.

This formulation naturally extends to K -layer encoders by computing path-wise weight products. For each path from input feature f through intermediate hidden layers to the latent space, we multiply the absolute weights along the path and sum over all possible paths:

$$I_f^{\text{weight}} = \sum_{\text{paths}} \prod_{\ell=1}^K |W_{\text{path}}^{(\ell)}|$$

Comment 37: The authors write “The difference in model performance was assessed as the absolute differences of the latent representation when changing each input to missing for all individuals and passing it through the trained model.” How did you in practice change the input to missing? It is not clear from the text.

Response: We thank you for this comment. We simulate the "missing" of a single feature——replace the entire column with its training mean in standardized space, which equals 0 after z-score normalization. This keeps perturbed inputs near the training distribution and yields stable, attribution-like importance scores.

Preprocess all continuous features once using training statistics (z-score normalization). Compute encoder means on the unperturbed design matrix X to obtain $\mu(X) \in \mathbb{R}^{N \times L}$. For each feature f , create a perturbed dataset $X^{(-f)}$ by setting column f to 0 (its standardized mean) for all individuals. Encode $X^{(-f)}$ to obtain $\mu(X^{(-f)})$. Define the perturbation score as the average absolute change in encoder means across individuals

and latent dimensions:

$$I_f^{\text{perturb}} = \sum_{j=1}^L \left(\frac{1}{N} \sum_{i=1}^N |\mu_j(x^{(i)}) - \mu_j(x^{(i,-f)})| \right)$$

Features are then ranked by I_f^{perturb} .

We have revised Methods section.

Comment 38: In the “Comparison for feature selection and model performance” section, it is described that a number of ML algorithms were compared using 10-fold CV, and the 10 clinical features. However, it is not described if/how models utilizing the full featureset perform.

Response: We thank you for this helpful comment and the opportunity to clarify this point.

As noted, the supervised machine learning analyses were not the primary focus of our study but were conducted to further validate the reliability of the features identified through the VAE in capturing T2D heterogeneity.

Specifically, we first applied multiple ML algorithms using the full clinical feature set to identify the best-performing classifiers. Gradient Boosting Classifier and Random Forest demonstrated the highest overall discriminative performance and were therefore selected for subsequent comparisons. We then evaluated their accuracy using the top 5-10 features derived from the VAE-based feature ranking. To address your question, we have now explicitly included the comparison between the full feature set and the VAE-selected subset in Supplementary Figure 5. The Gradient Boosting model achieved an AUC of 0.926 using all features, which remained comparable (AUC = 0.916) when restricted to the top 10 variables, with similarly high PR-AUC and F1 scores. These findings confirm that the ten selected features are sufficient to capture the major discriminative patterns in our dataset.

We have clarified this analytical workflow and rationale in the revised Methods section to improve transparency.

Comment 39: In Figure 6, panel f, a ribbon chart is showing transitions between groups. These groups appear on the right in the figure, but it is not specified how these were defined. Also, there are some individuals not belonging to a group – is this intended? The methods section lacks details on how these were defined.

Response: We thank you for this comment. We clarify that the seven groups were manually defined based on approximate regions of the Chinese tree coordinates to facilitate visualization of potential 5-year trajectories. Because DDRTree produces a continuous representation rather than hard clusters, some individuals naturally fall outside these predefined groups. This approach is intended to illustrate trends in subgroup trajectories rather than to assign individuals to rigid clusters. We have added the details in the figure legend.

Minor comments

Comment 40: in the abstract, there is a missing “the” in the sentence “built a tree structure using Discriminative Dimensionality Reduction Tree (DDRTree) algorithm.”

Response: We thank you for pointing this out. The sentence has been corrected to: built a tree structure using **the** Discriminative Dimensionality Reduction Tree (DDRTree) algorithm.

Comment 41: The authors write “In the Chinese tree, 6.45% (95 out of 1,472, from 319 26,000 Chinese individuals) showed a shift (distance > 1) from their baseline position.”

How is this distance calculated?

Response: We thank you for the comment. The reported distance represents the Euclidean distance between points in the two-dimensional Chinese tree space. Distances were calculated using a distance-estimating algorithm applied to the DDRTree coordinates of each individual at baseline and follow-up.

Comment 42: In the discussion, the authors write: “This continuous model structure may offer higher clinical relevance than rigid cluster boundaries, especially for diseases

like T2D where pathophysiology lies on a continuum.” There are ways to enforce non-rigid cluster boundaries in cluster algorithms – see e.g. fuzzy clustering methods or probabilistic clustering.

Response: We thank you for this comment. We have revised the text to clarify that the DDRTree provides a continuous representation of individuals’ heterogeneity. This continuous model structure may better capture gradual transitions between phenotypes and reflect the spectrum of T2D heterogeneity.

Comment 43: In lines 600-601 you write first decoder and then encoder. I suppose the encoder should be changed to decoder?

Response: We thank you for pointing this out. The text has been corrected to accurately describe the VAE architecture, ensuring that “decoder” and “encoder” are used in the proper context.

Comment 44: There is an error in the loss formula – MCE is written instead of MSE.

Response: We thank you for pointing this out. The text has been corrected to MSE.

Comment 45: There are repeat sentences in lines 630-632.

Response: We thank you for pointing this out. The sentence has been corrected.

REVIEWERS' COMMENTS

Reviewer #1 (Remarks to the Author):

The authors have addressed my previous comments. My only remaining comments are minor and consist of typos I identified while going through the new manuscript.

Response: We sincerely thank the reviewer for the careful re-reading of our revised manuscript and for identifying these minor typographical issues. We appreciate the reviewer's constructive comments and careful attention, which have helped further improve the clarity and accuracy of our manuscript.

Specifically:

1) In figure 1 and the panel of step 2, the word “hyperparamter” is written instead of hyperparameter.

Response: We have corrected the typo “hyperparamter” to “hyperparameter” in the **Figure 1 (Step 2 panel)**.

2) In line 672, it reads: “where corresponds to the inverse of C”. I believe that this should read “which corresponds to the inverse of C”.

Response: We have revised the sentence from “where corresponds to the inverse of C” to “which corresponds to the inverse of C” as suggested.